Dimensionality reduction methods (PCA / PLSDA / MDS)

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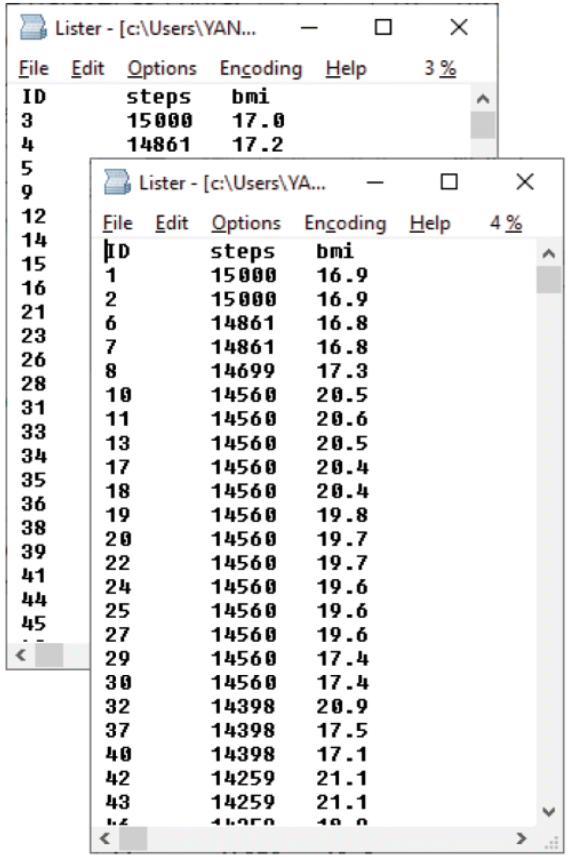
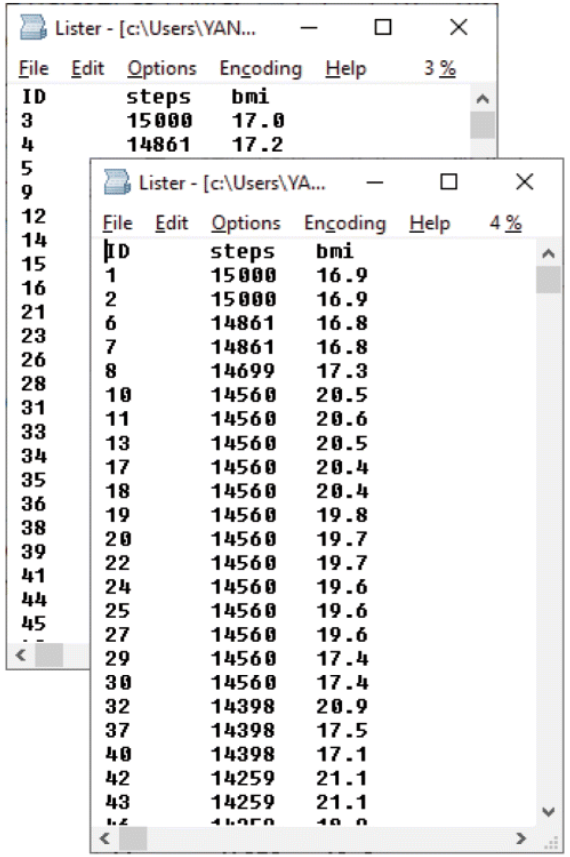
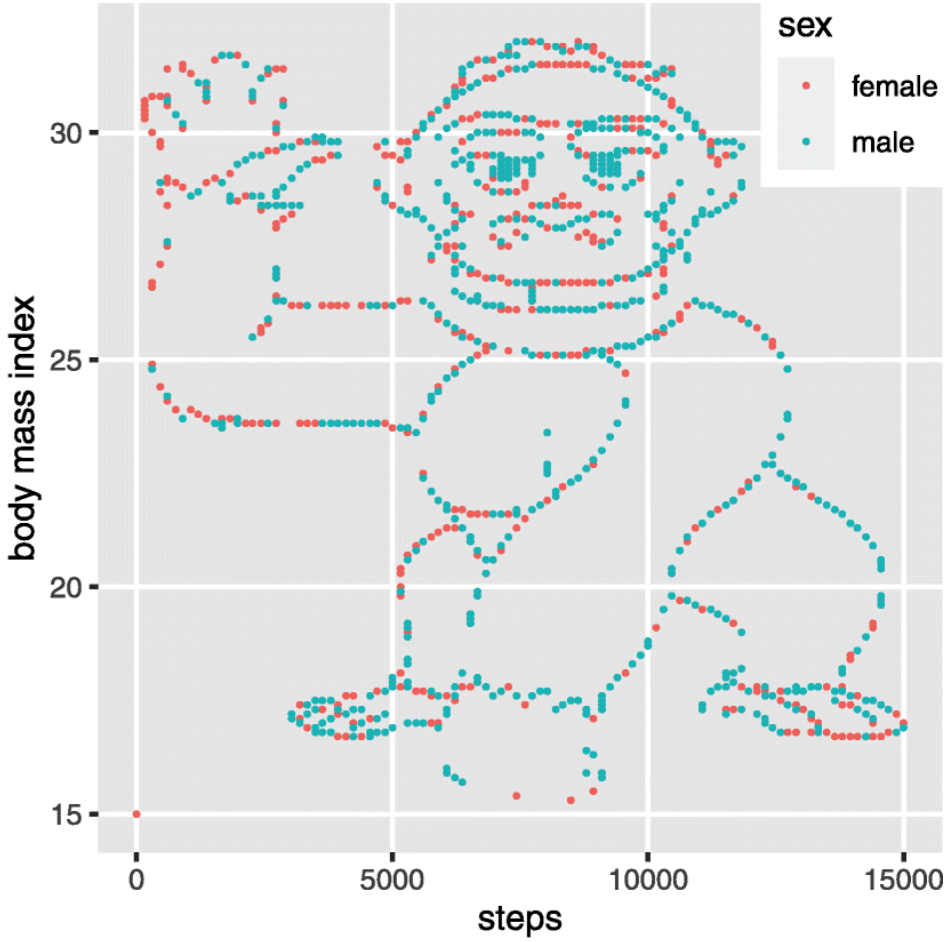
Hypothesis as a liability



Yanai & Lercher (BMC Genome Biology 2020)

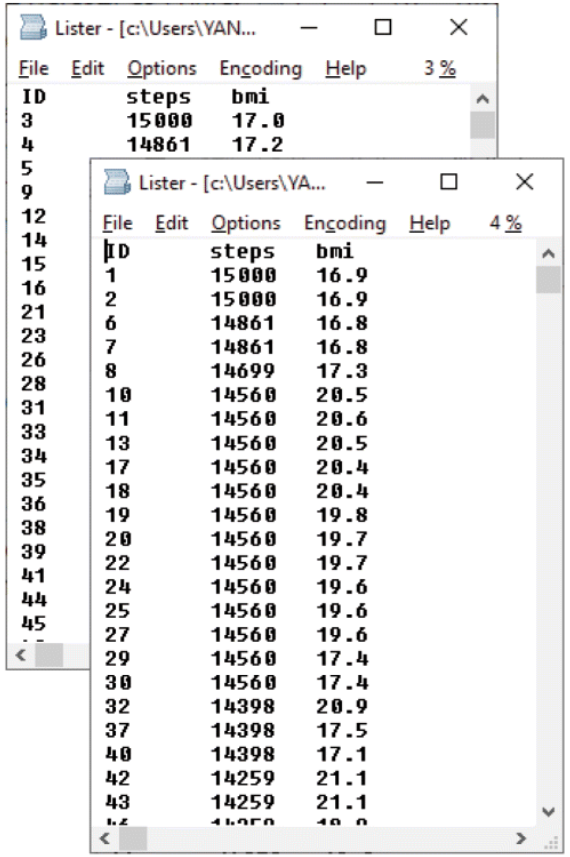
Hypothesis as a liability

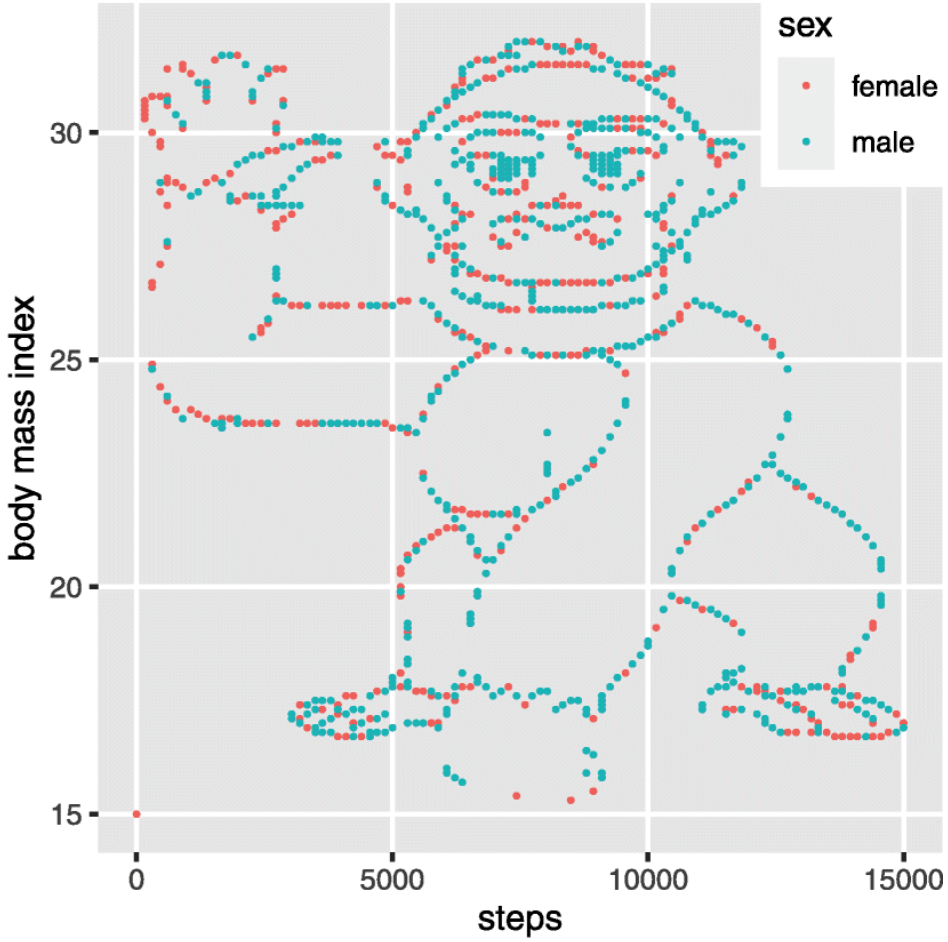




Yanai & Lercher (BMC Genome Biology 2020)

Hypothesis as a liability





Hypothesis	🚫	✅
✅	14	5
🚫	5	9

Yanai & Lercher (BMC Genome Biology 2020)

Numbers don’t tell the whole story

A		B		C		D	
x	y	x	y	x	y	x	y
10.0	8.04	10.0	9.14	10.0	7.46	8.0	6.58
8.0	6.95	8.0	8.14	8.0	6.77	8.0	5.76
13.0	7.58	13.0	8.74	13.0	12.74	8.0	7.71
9.0	8.81	9.0	8.77	9.0	7.11	8.0	8.84
11.0	8.33	11.0	9.26	11.0	7.81	8.0	8.47
14.0	9.96	14.0	8.10	14.0	8.84	8.0	7.04
6.0	7.24	6.0	6.13	6.0	6.08	8.0	5.25
4.0	4.26	4.0	3.10	4.0	5.39	19.0	12.50
12.0	10.84	12.0	9.13	12.0	8.15	8.0	5.56
7.0	4.82	7.0	7.26	7.0	6.42	8.0	7.91
5.0	5.68	5.0	4.74	5.0	5.73	8.0	6.89

Numbers don’t tell the whole story

A		B		C		D	
x	y	x	y	x	y	x	y
10.0	8.04	10.0	9.14	10.0	7.46	8.0	6.58
8.0	6.95	8.0	8.14	8.0	6.77	8.0	5.76
13.0	7.58	13.0	8.74	13.0	12.74	8.0	7.71
9.0	8.81	9.0	8.77	9.0	7.11	8.0	8.84
11.0	8.33	11.0	9.26	11.0	7.81	8.0	8.47
14.0	9.96	14.0	8.10	14.0	8.84	8.0	7.04
6.0	7.24	6.0	6.13	6.0	6.08	8.0	5.25
4.0	4.26	4.0	3.10	4.0	5.39	19.0	12.50
12.0	10.84	12.0	9.13	12.0	8.15	8.0	5.56
7.0	4.82	7.0	7.26	7.0	6.42	8.0	7.91
5.0	5.68	5.0	4.74	5.0	5.73	8.0	6.89

Summary statistics

Mean(x):	A=B=C=D=9	exact
Variance(x):	A=B=C=D=11	exact
Mean(y):	A=B=C=D=7.50	(2 decimals)
Variance(y):	A=B=C=D=4.125	(+/- 0.003)
Correlation(x,y):	A=B=C=D=0.816	(3 decimals)

Numbers don’t tell the whole story

A		B		C		D	
x	y	x	y	x	y	x	y
10.0	8.04	10.0	9.14	10.0	7.46	8.0	6.58
8.0	6.95	8.0	8.14	8.0	6.77	8.0	5.76
13.0	7.58	13.0	8.74	13.0	12.74	8.0	7.71
9.0	8.81	9.0	8.77	9.0	7.11	8.0	8.84
11.0	8.33	11.0	9.26	11.0	7.81	8.0	8.47
14.0	9.96	14.0	8.10	14.0	8.84	8.0	7.04
6.0	7.24	6.0	6.13	6.0	6.08	8.0	5.25
4.0	4.26	4.0	3.10	4.0	5.39	19.0	12.50
12.0	10.84	12.0	9.13	12.0	8.15	8.0	5.56
7.0	4.82	7.0	7.26	7.0	6.42	8.0	7.91
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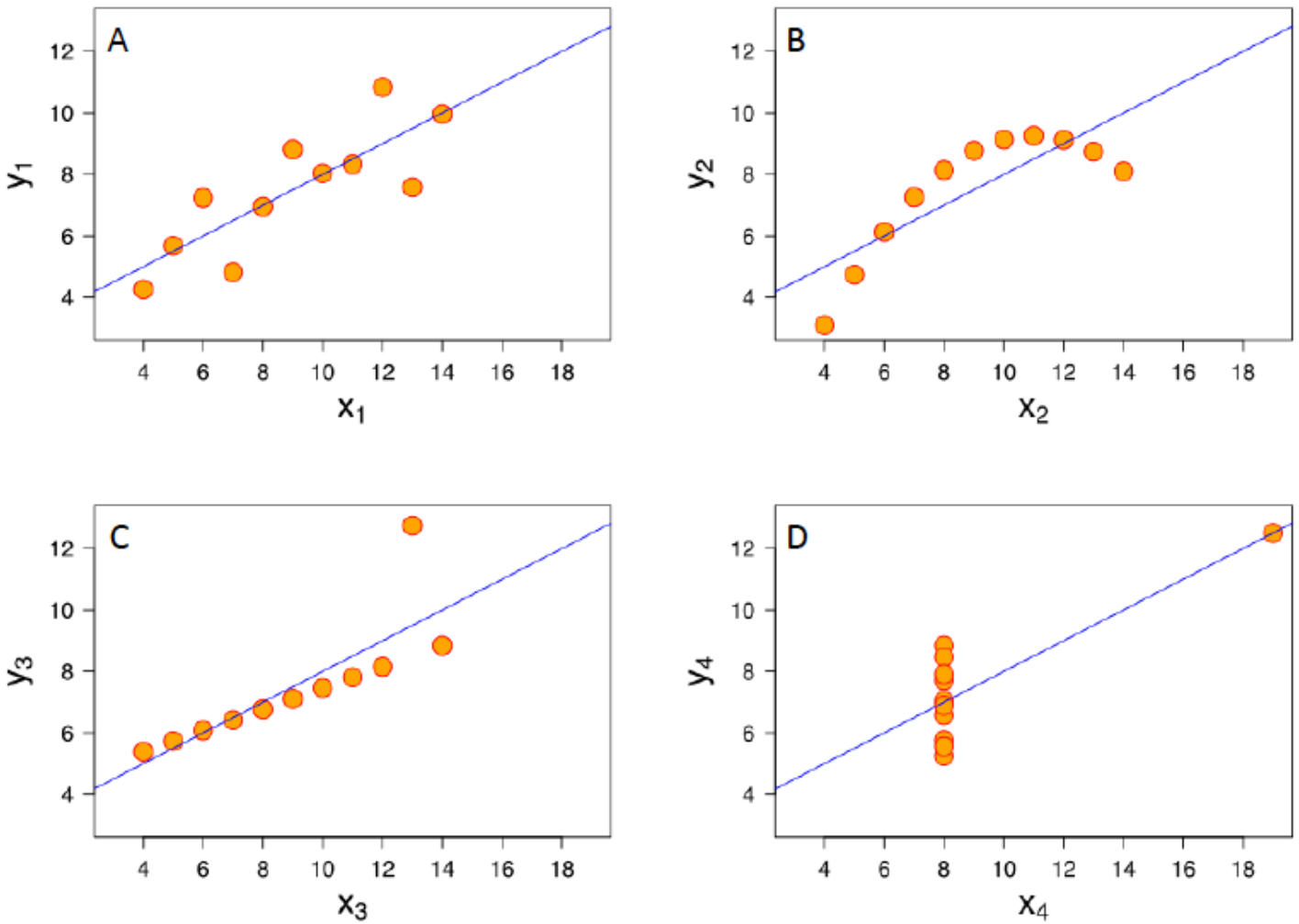
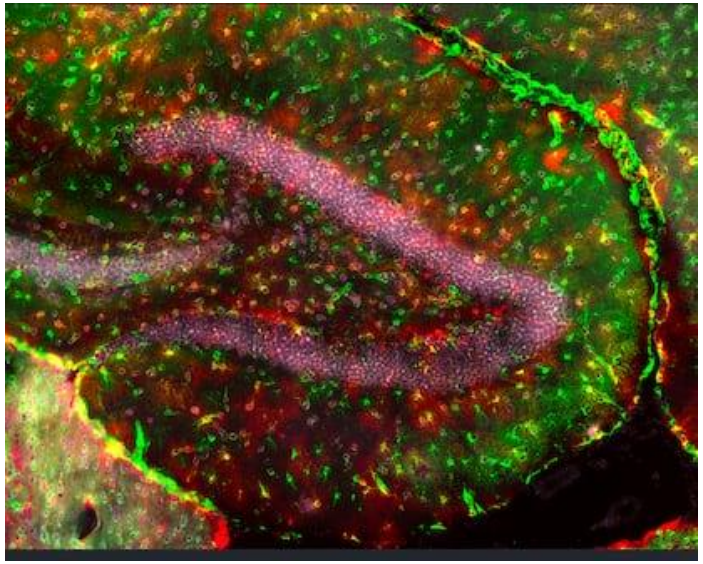


Anscombe’s Quartet, 1973

Biology

Discover new knowledge



Machine learning / Statistics

Predict existing phenomena / test a priori defined hypothesis

How can we visualize high-dimensional data

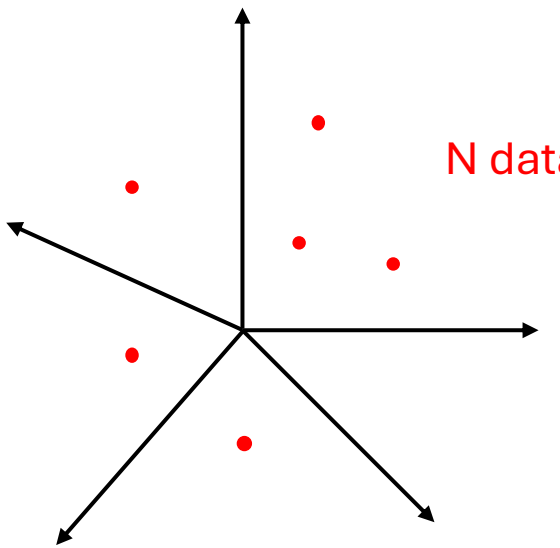
Features: genes, proteins, microbes, metabolites etc. ← P

N

Statistical observations: e.g. samples, cells etc.

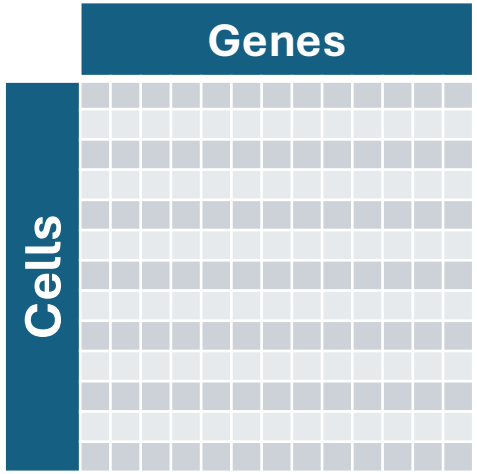
$$\begin{bmatrix} 0 & 8 & 34 & 85 & 44 \\ 1 & 4 & 6 & 35 & 31 \\ 8 & 6 & 78 & 8 & 92 \\ 10 & 3 & 23 & 5 & 3 \\ 6 & 45 & 98 & 76 & 8 \\ 32 & 8 & 23 & 84 & 0 \end{bmatrix}$$

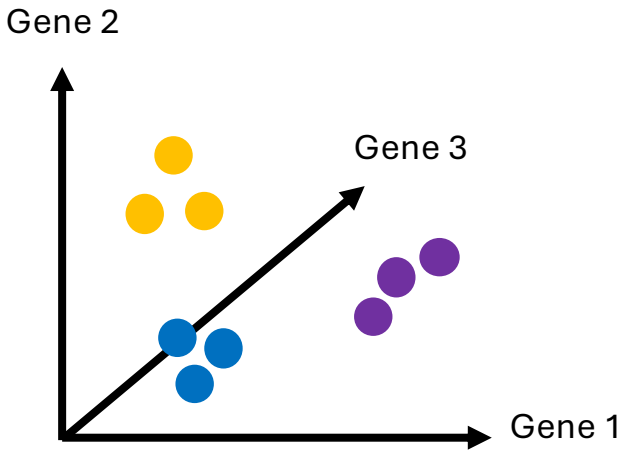
P-dimensional space

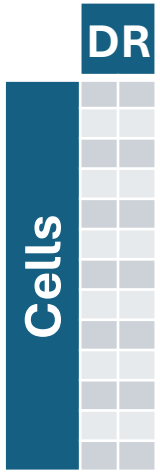


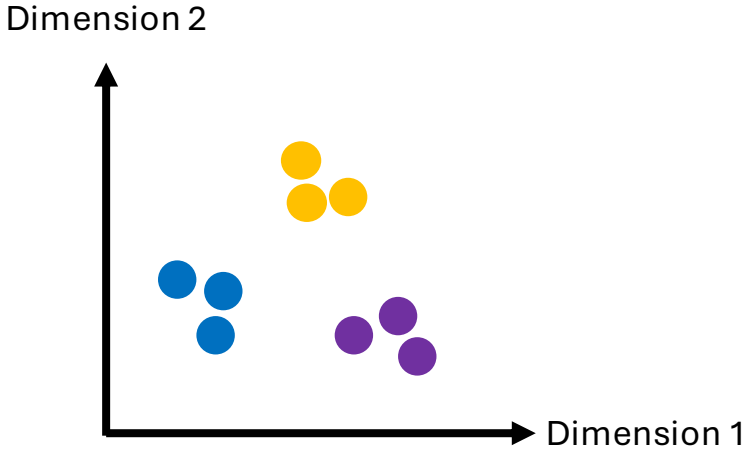
N data points

Dimensionality reduction







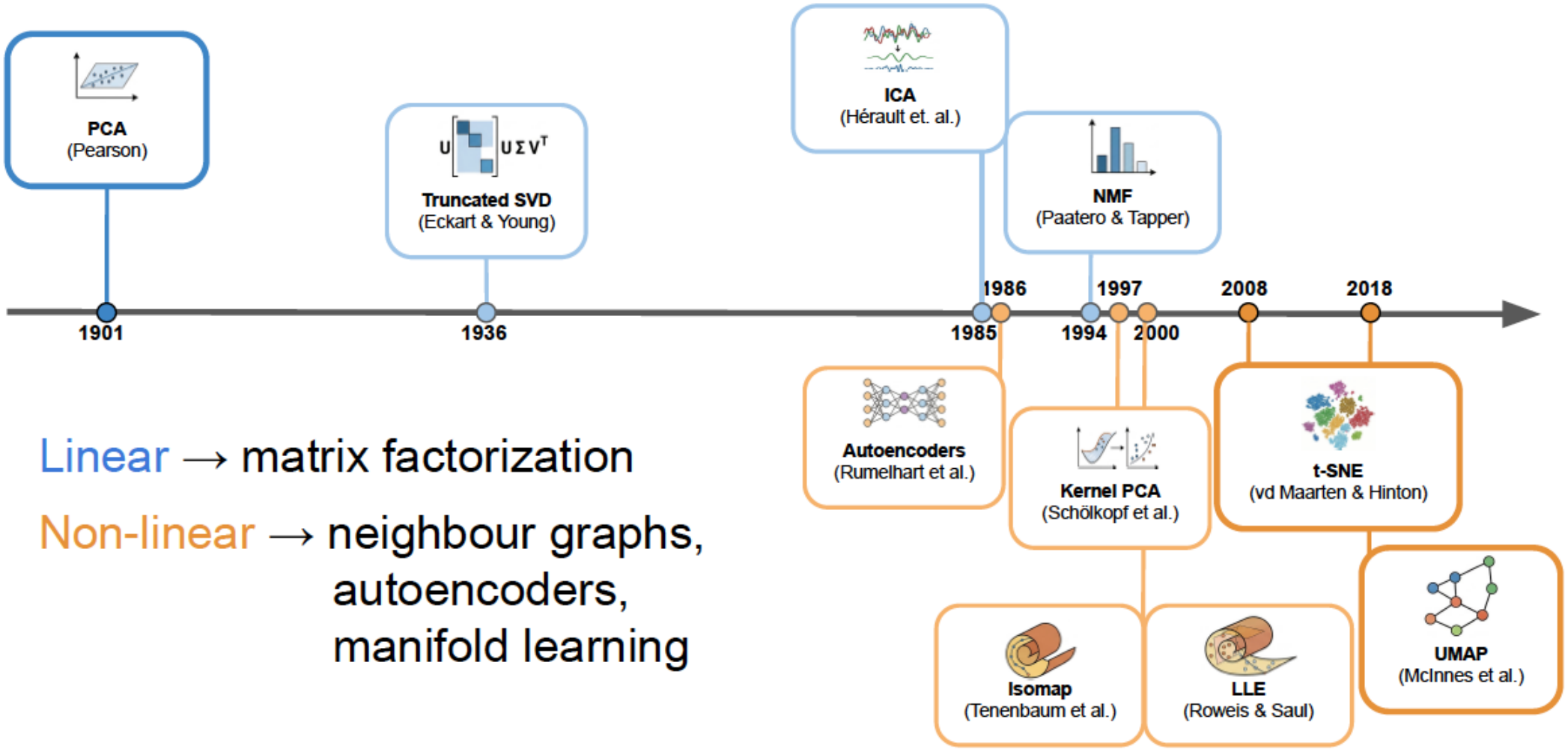


Why dimensionality reduction?

- Visualization

- We want to reduce complexity for analysis
  - “Remove” redundancies in the data
  - Identify the most relevant information (find and filter noise)
  - Reduce computational time for downstream procedures e.g. clustering

Dimensionality reduction methods

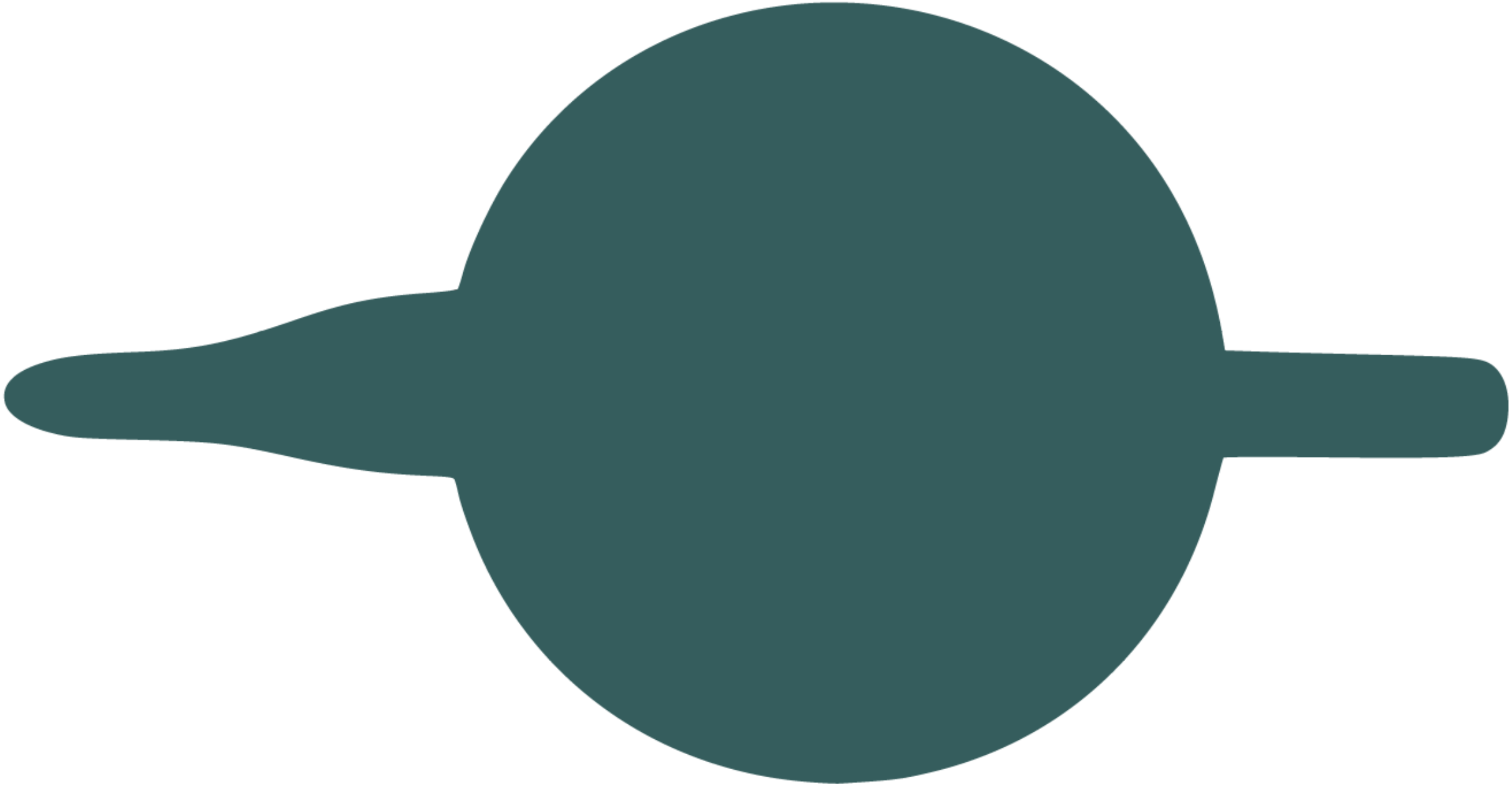


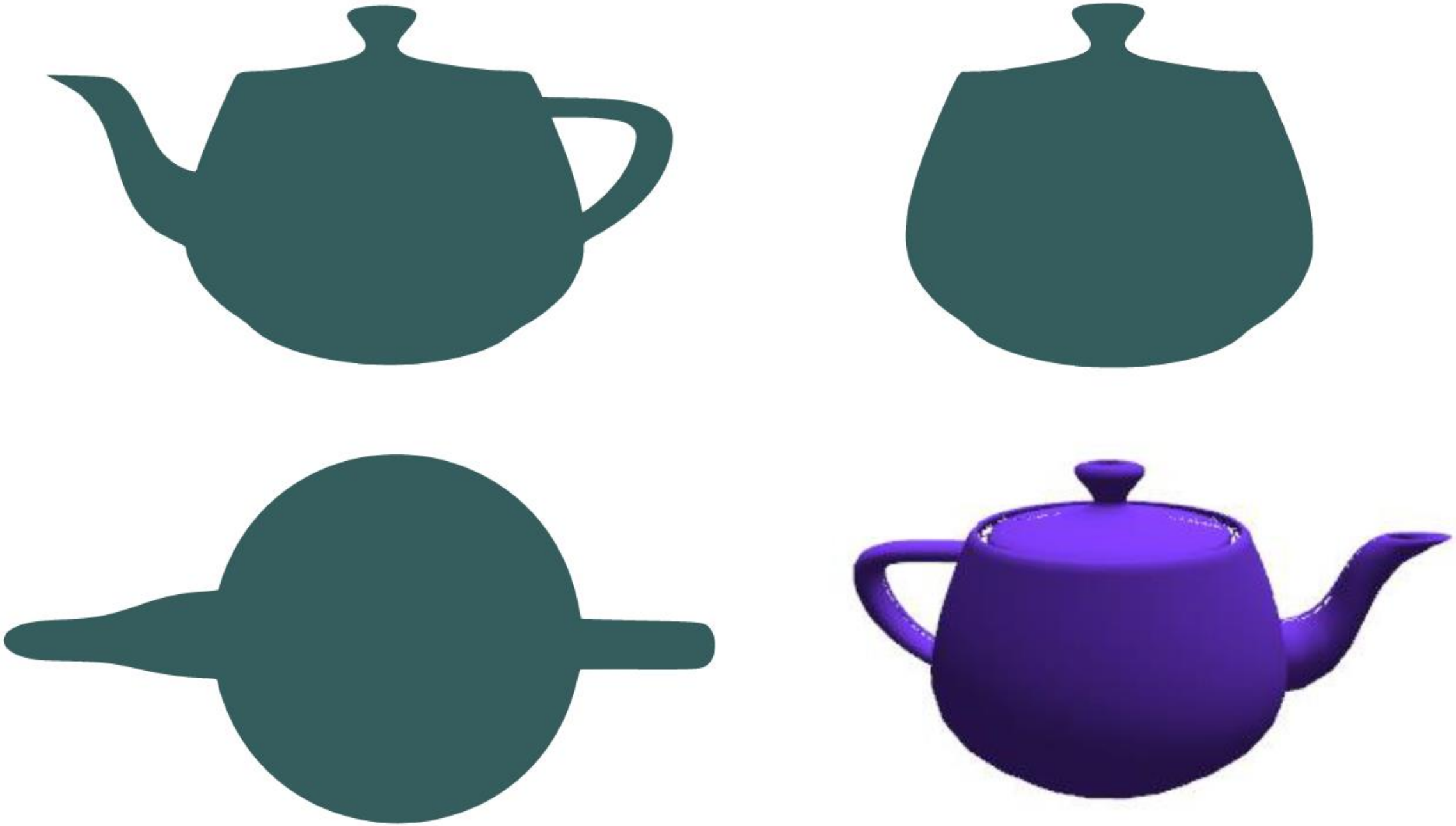
Linear → matrix factorization

Non-linear → neighbour graphs,
autoencoders,
manifold learning

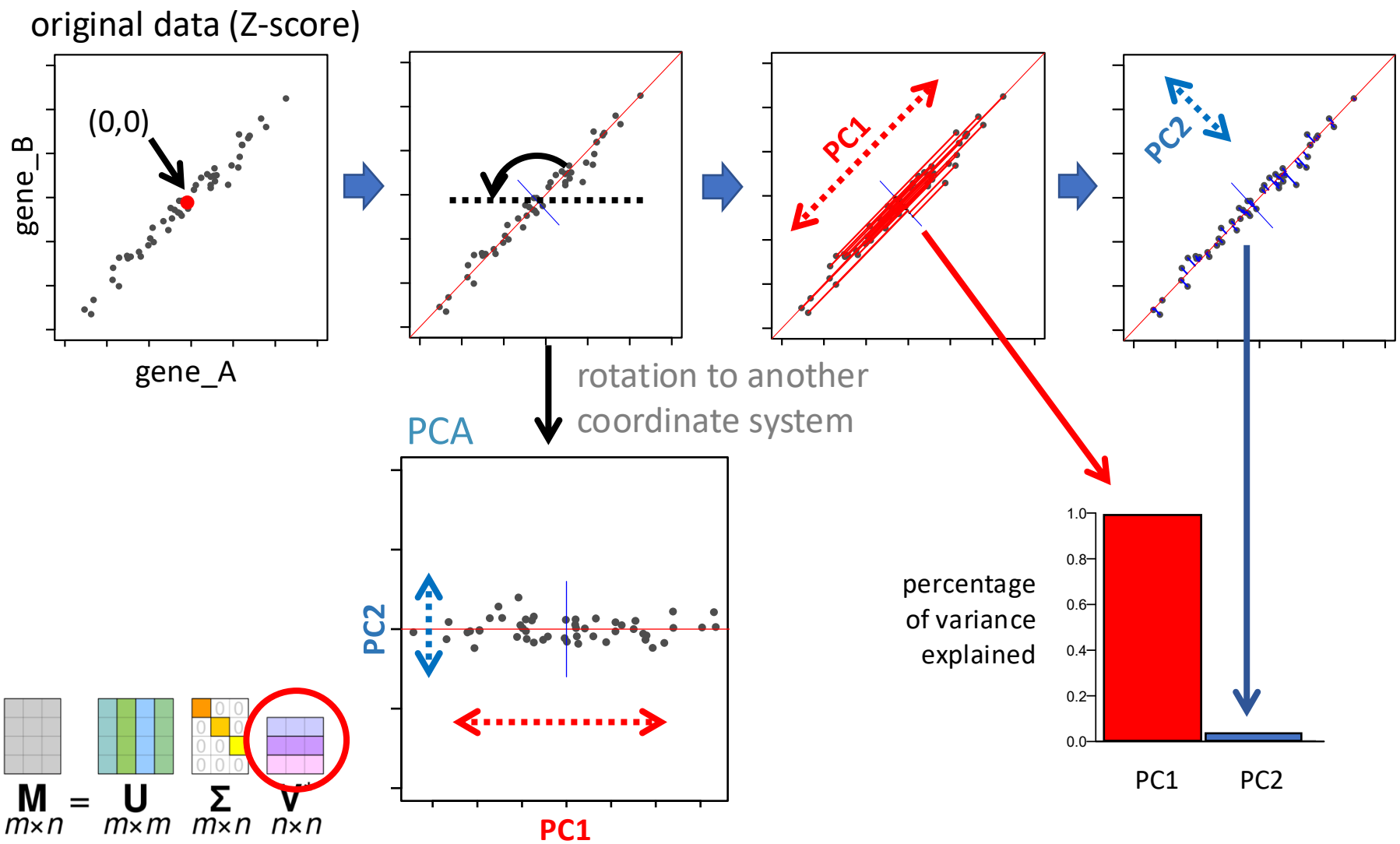
Slide courtesy Susanne Reinsbach

Principal **C**omponent **A**nalysis (PCA)



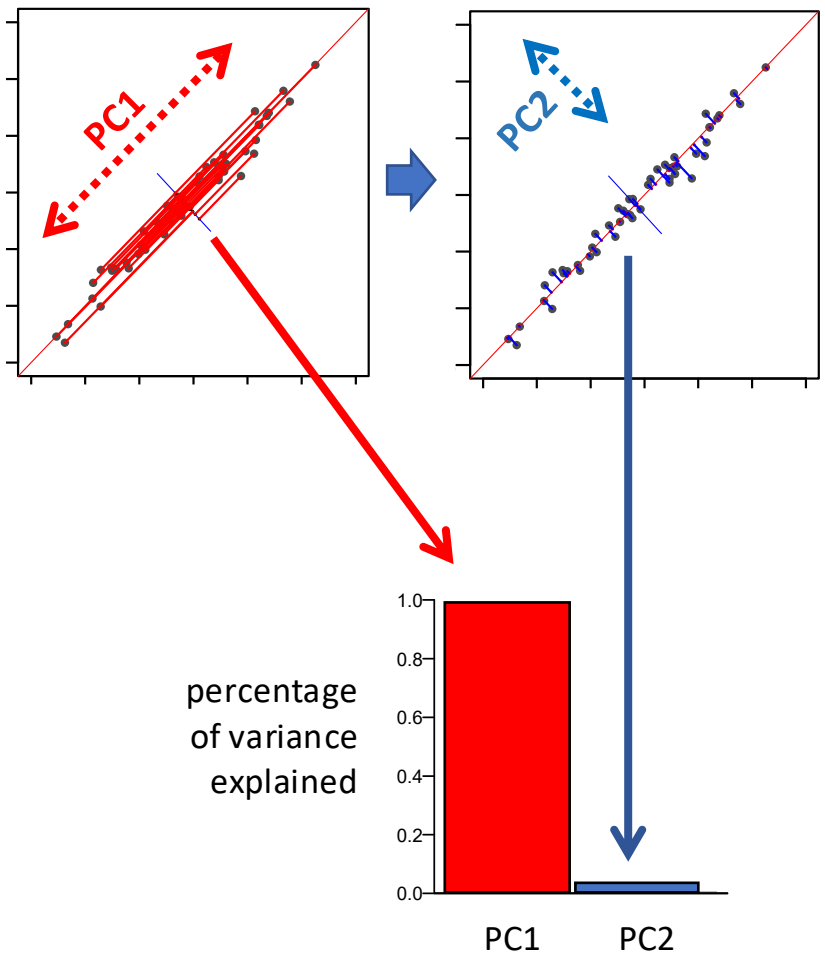


Principle Component Analysis (PCA)



Principle Component Analysis (PCA)

- PC1 explains >98% of the variance
- 1 PC thus represents 2 genes very well
  - “Removing” redundancy
- PC2 is nearly insignificant in this example
  - Could be disregarded



Principle Component Analysis (PCA)

$X = N \times P$

	Var 1	Var 2
1	3.20	0.07
2	1.65	0.07
3	3.28	0.12
4	1.01	0.19
5	2.71	0.24
6	2.06	0.08
7	1.85	0.12
8	2.71	0.18
9	2.36	0.08
10	1.93	0.25

$0.2 \times$

Var 1
3.20
1.65
3.28
1.01
2.71
2.06
1.85
2.71
2.36
1.93

$+ \quad 0.8 \times$

Var 2
0.07
0.07
0.12
0.19
0.24
0.08
0.12
0.18
0.08
0.25

$=$

Linear combination
0.69
0.39
0.75
0.35
0.73
0.48
0.47
0.69
0.53
0.59

- Linear combination (component) summarizes the original data X
- The weights (loadings) are set to optimize a statistical criterion (maximize variance)
- Solved with Singular Value Decomposition algorithm for large data sets

PCA for higher dimensional data

$data = N \times P$

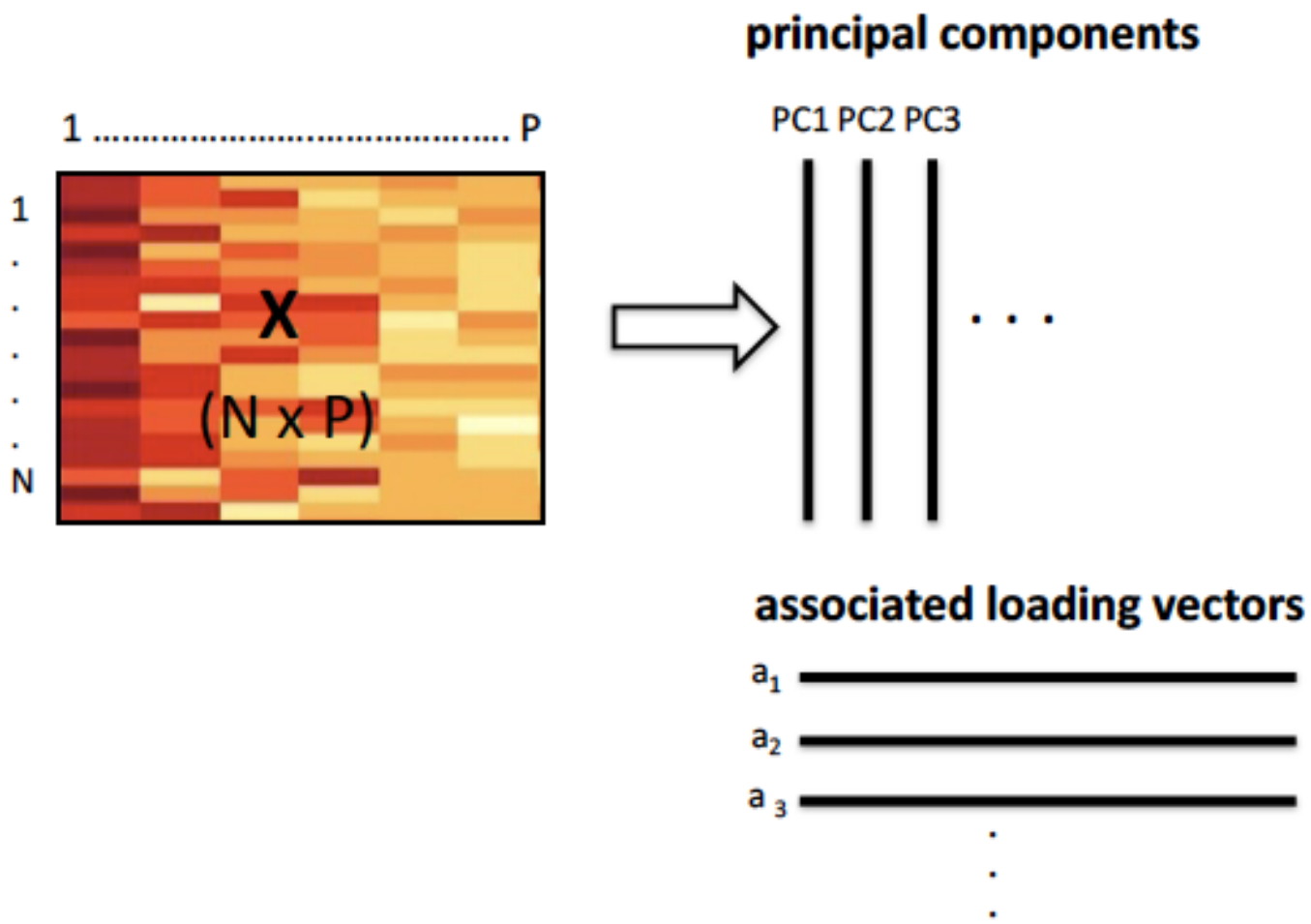
	Var 1	Var 2	Var 3	Var4	Var5
1	3.97	3.16	3.54	3.89	2.11
2	2.05	2.36	3.89	3.50	1.58
3	3.36	3.11	4.20	2.26	-0.09
4	4.15	4.79	5.43	3.30	2.00
5	2.91	3.05	3.87	4.99	1.69
6	3.44	3.96	3.54	3.89	2.11
7	3.65	2.22	3.89	3.50	1.58
8	4.40	3.60	4.20	2.26	-0.09
9	2.68	2.29	5.43	3.30	2.00
10	3.85	3.89	3.87	4.99	1.69

$components = N \times H$

	Comp1	Comp2	Comp3	Comp4	Comp5
1	0.62	0.31	-0.45	0.71	-0.15
2	0.41	-1.53	0.05	-0.34	-0.47
3	-1.91	-0.53	-0.35	-0.39	-0.07
4	-0.33	1.72	1.31	-0.08	-0.11
5	1.36	-0.24	-0.38	-0.52	0.35
6	0.69	0.64	-0.29	0.02	-0.69
7	0.07	-0.71	-0.29	0.73	0.26
8	-2.16	0.46	-0.53	0.04	0.18
9	0.14	-1.10	1.45	0.13	0.27
10	1.11	0.97	-0.52	-0.31	0.42

- Components are orthogonal to ensure no overlap of information
- Not all components are needed to summarize the information
  (H << P) → data compression
- Loading vectors indicate the weight (importance) of each variable in the linear combination

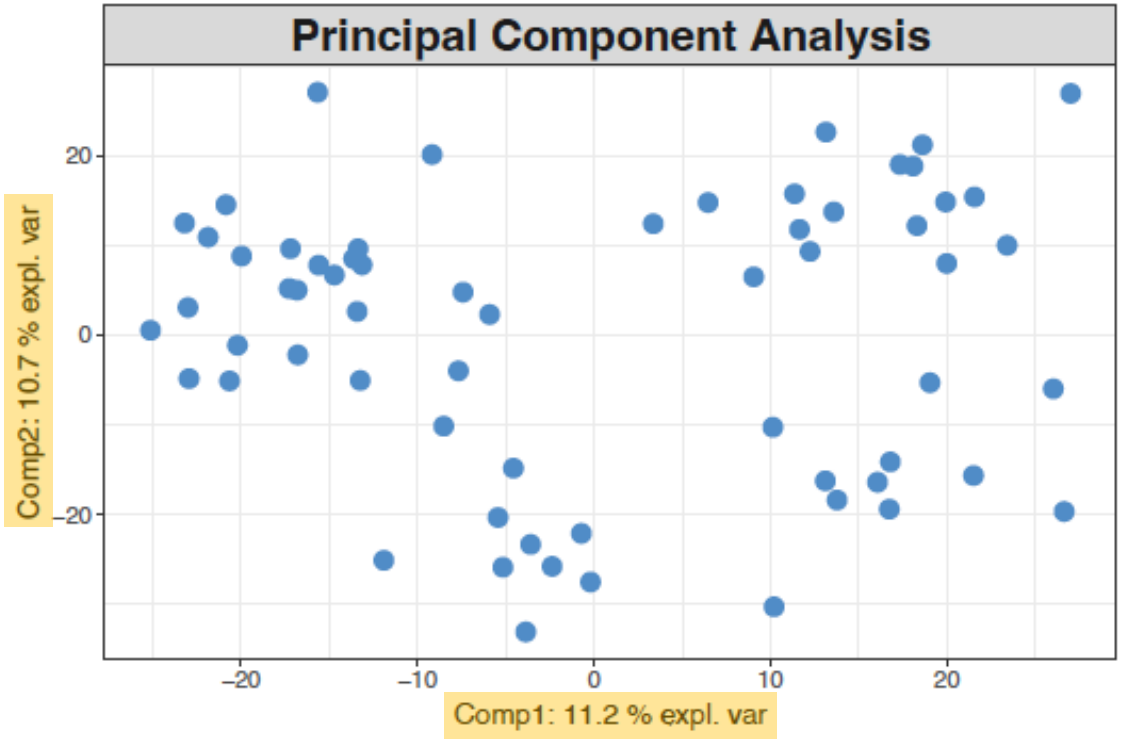
PCA is a matrix decomposition method



PCA for visualization

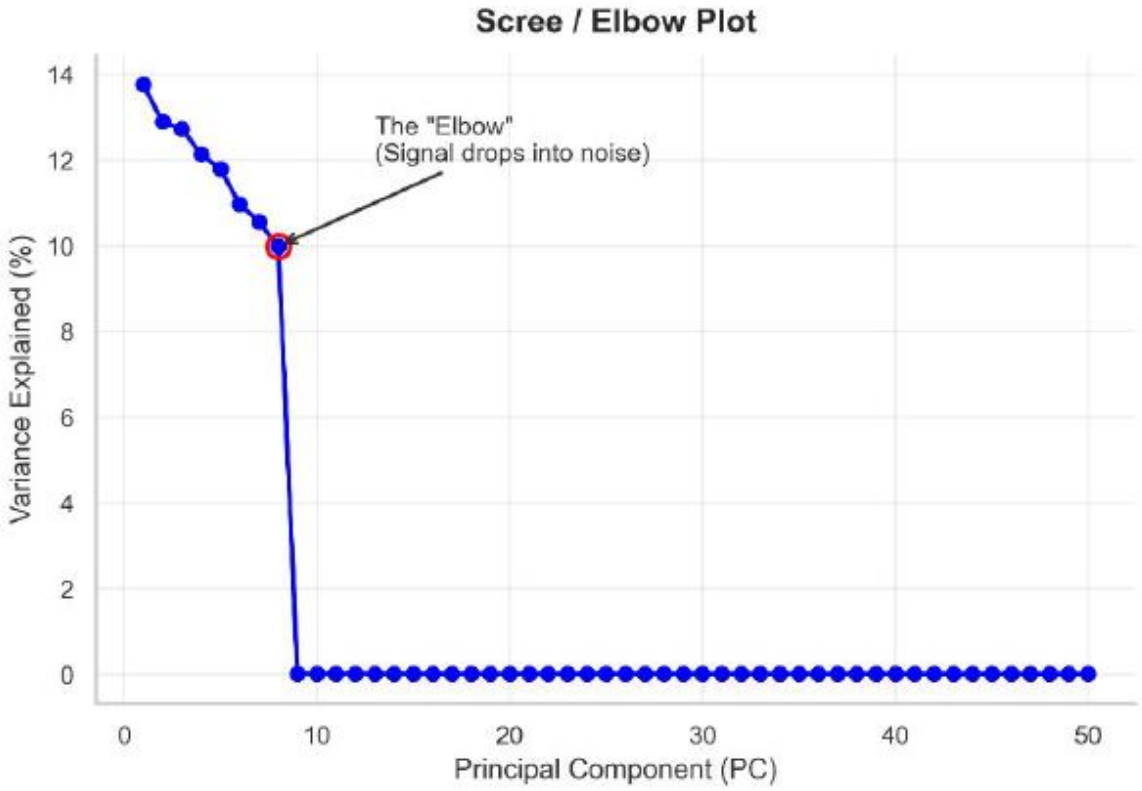
- Visualization: project data into a small subspace spanned by the components

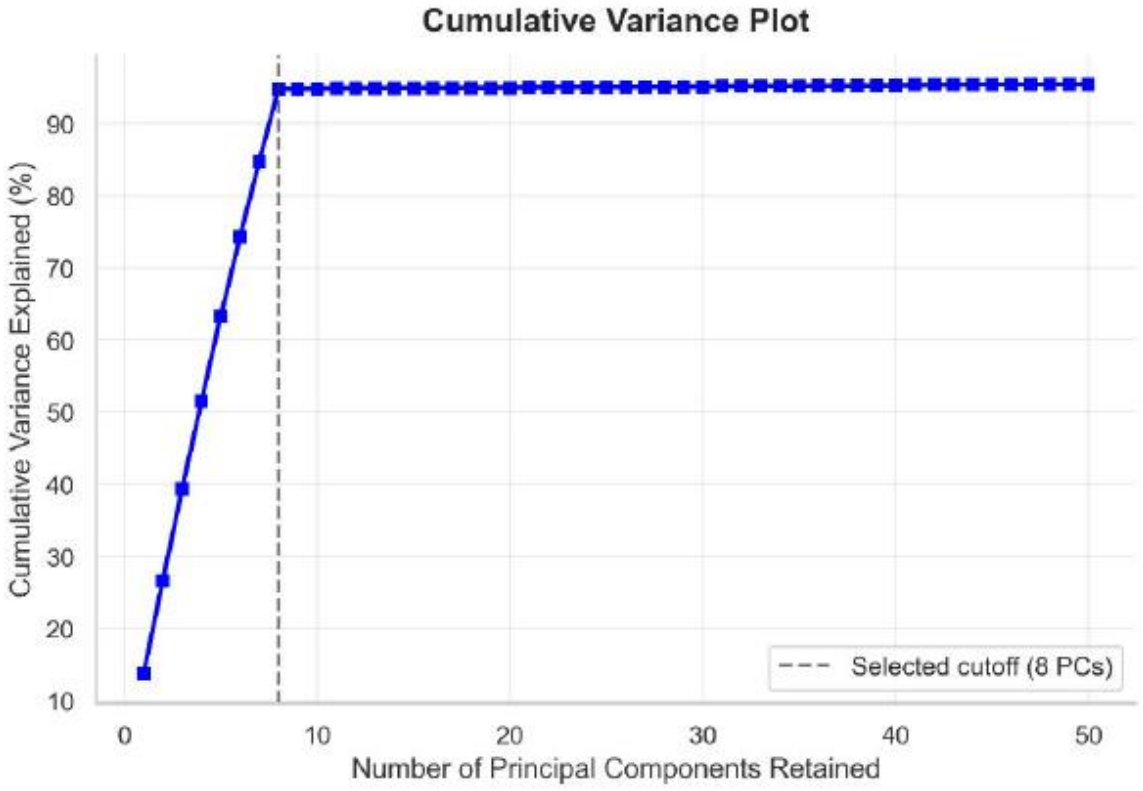
- ‘Similar’ samples (based on their variable values) are closer to each other



Visualization of 63 samples x 2,300 variables (genes)

Useful visualizations



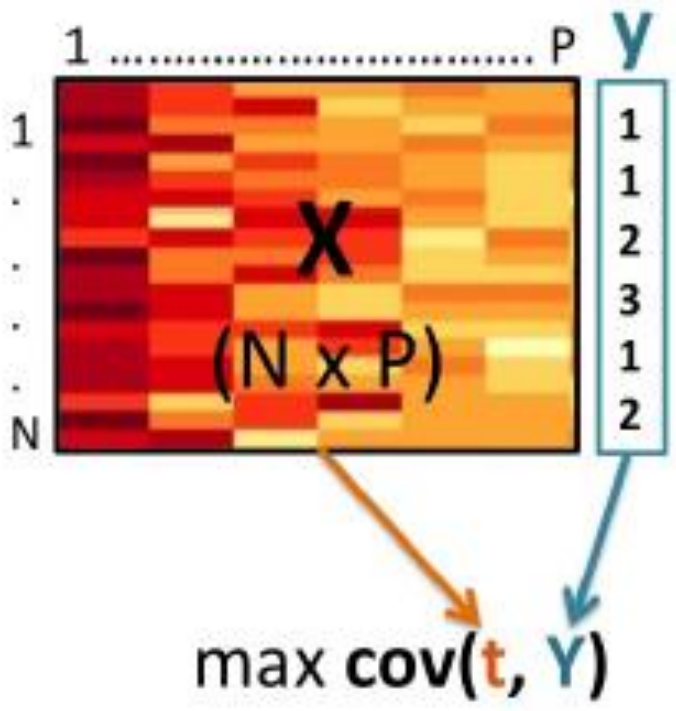


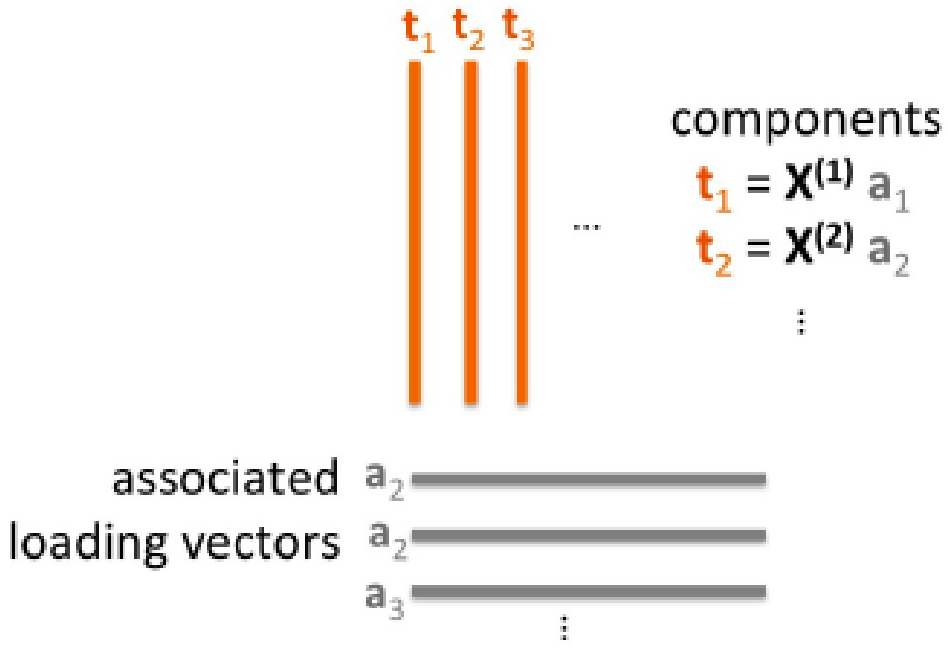
Slide courtesy Susanne Reinsbach

Projection to **L**atent **S**tructure **D**iscriminant **A**nalysis (PLS-DA)

PLS-DA

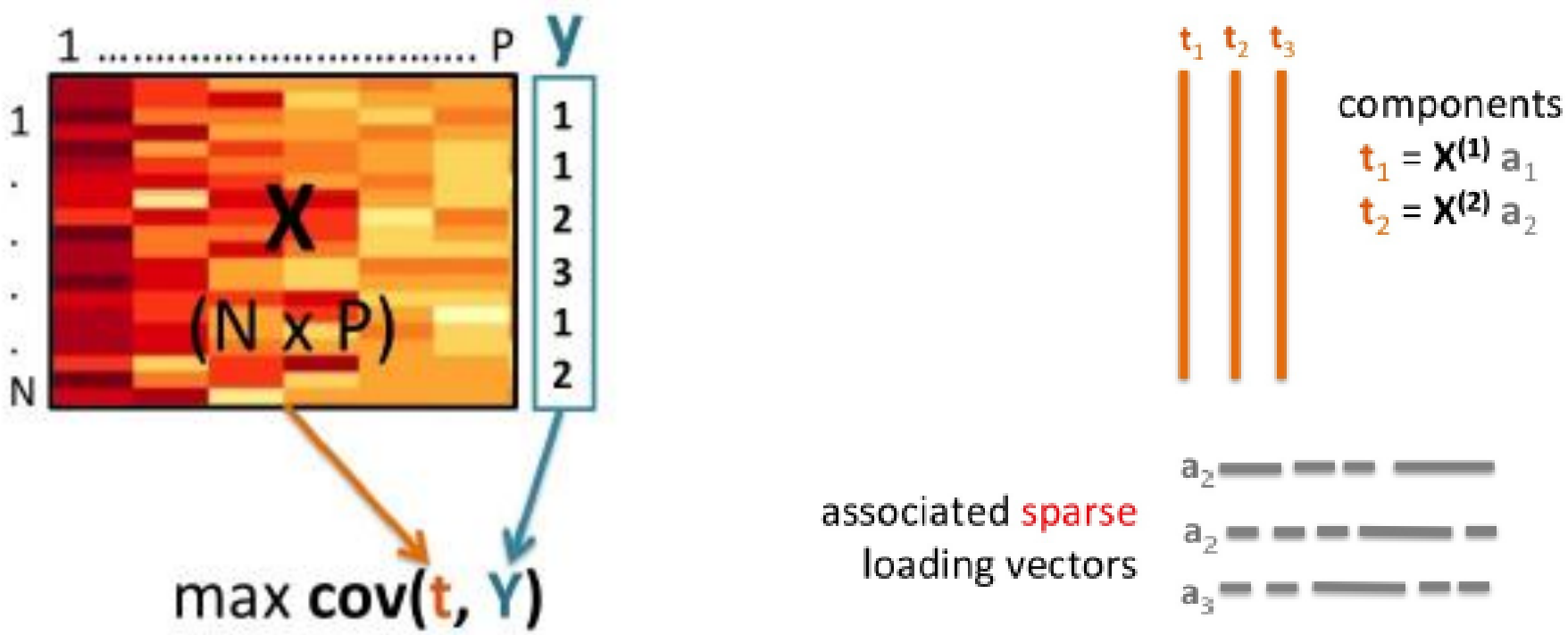
- What if we have an outcome that we are interested in?
- Can we guide PCA to maximize variation with respect to that outcome?





Sparse PLS-DA?

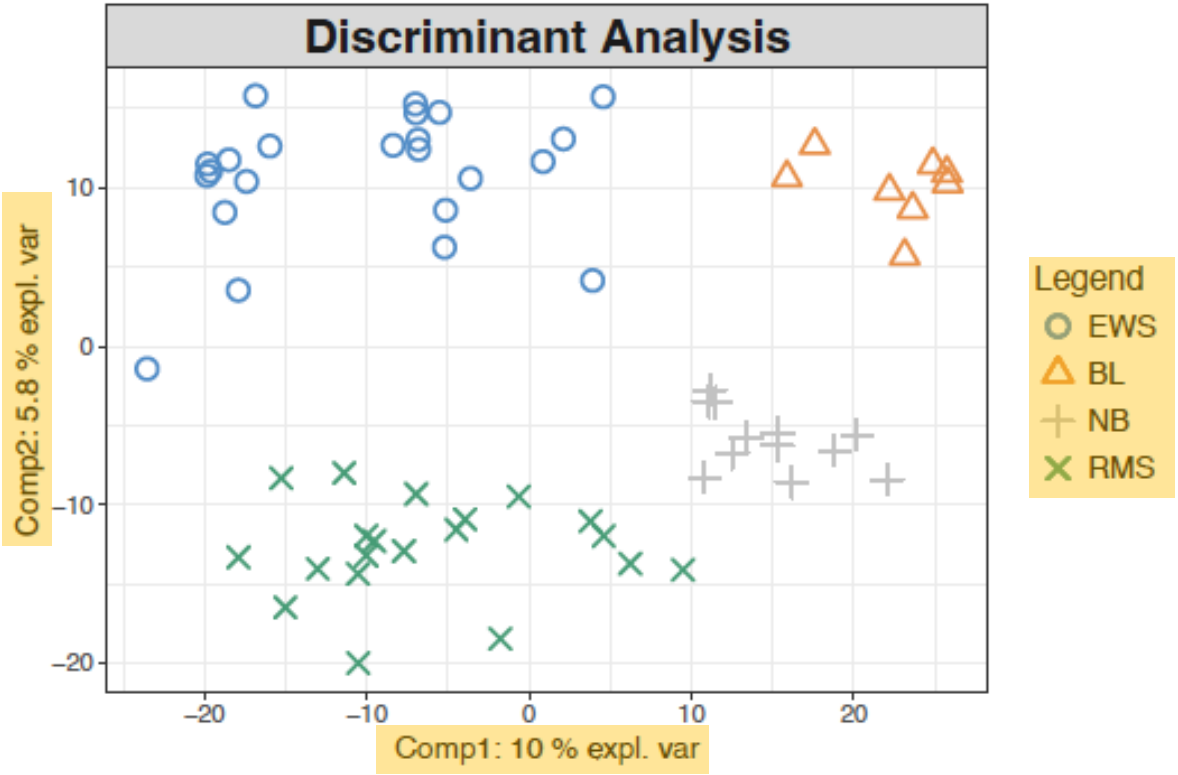
- Set some variable coefficients to zero (variable selection) for each component
- Each component is built on its selected variables only



Lê Cao et al. (2011). Sparse PLS Discriminant Analysis. BMC Bioinformatics 12:253.

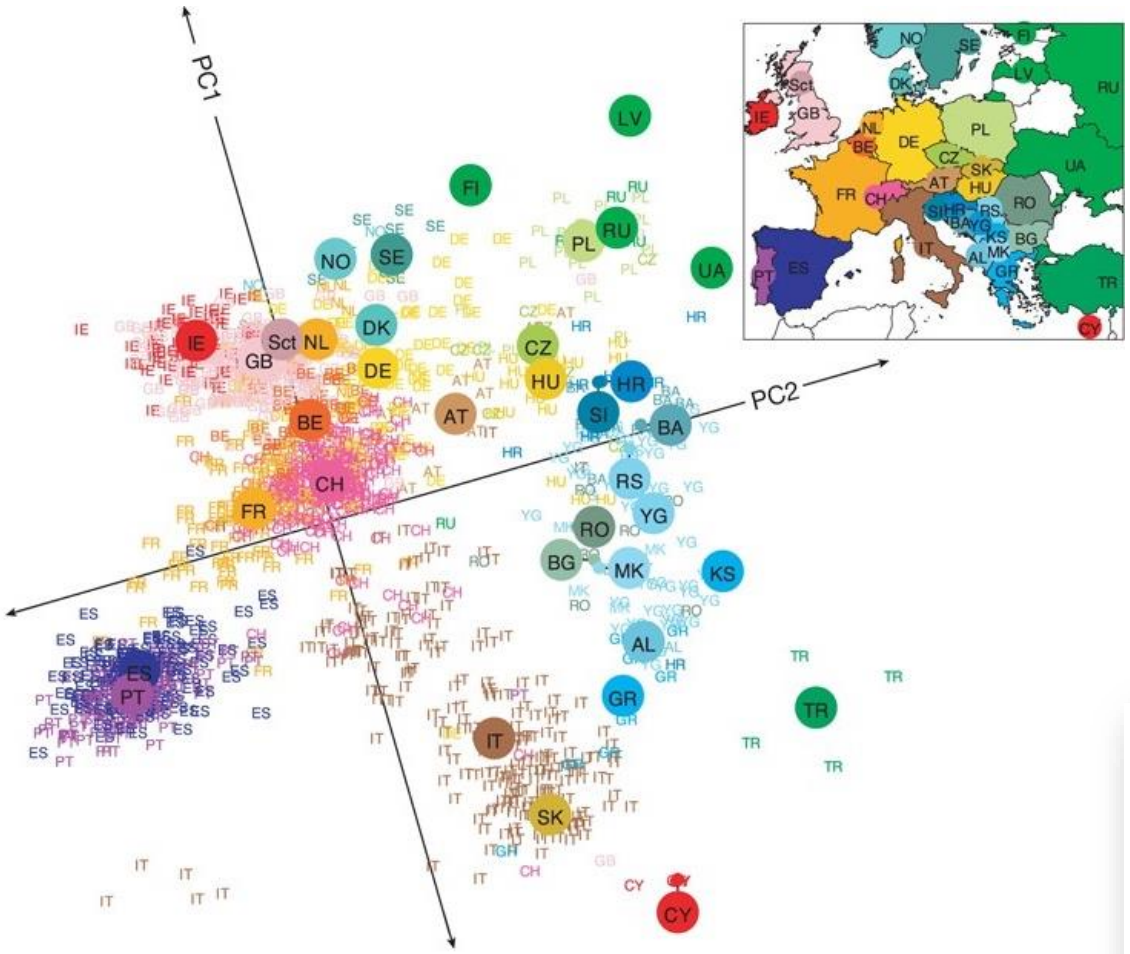
PLS-DA for visualization

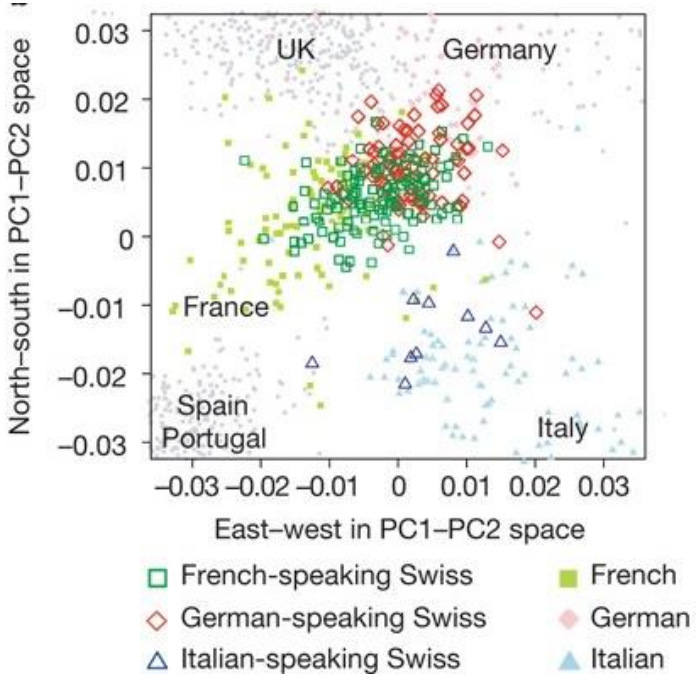
- Samples cluster according to their group
- Supervised analysis: the aim is to separate/discriminate sample

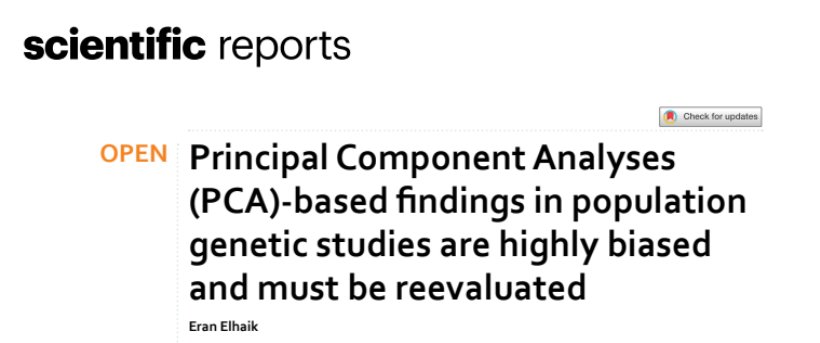


Visualization of 63 samples x 2,300 genes according to their tumor subtype

Pitfalls of PCA






scientific reports

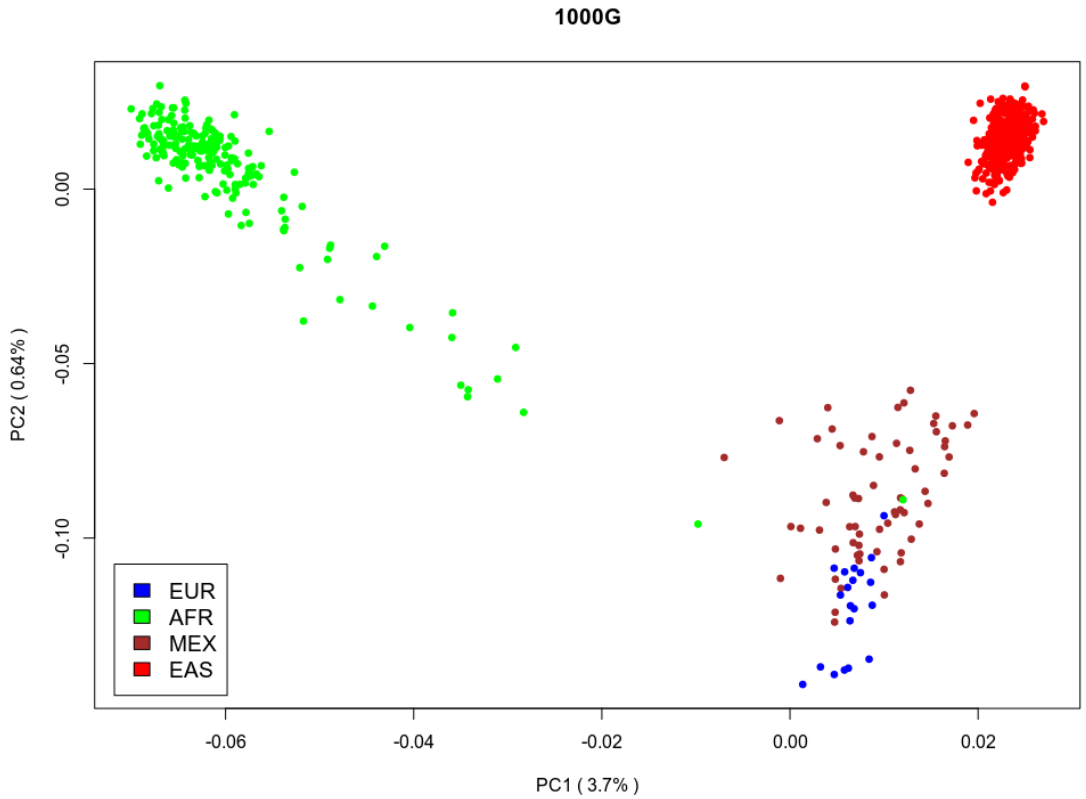
Check for updates

OPEN **Principal Component Analyses (PCA)-based findings in population genetic studies are highly biased and must be reevaluated**

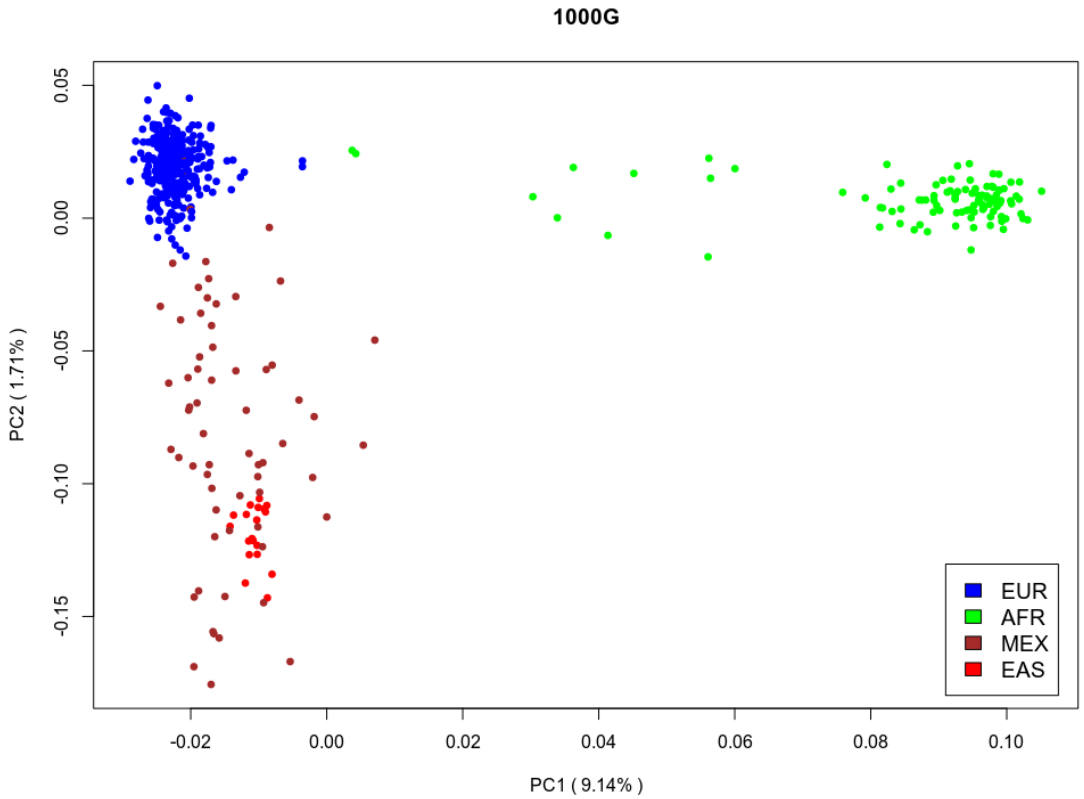
Eran Elhaik

Novembre et al. Nature 2008

Pitfalls of PCA



Downsampled Europeans (blue)



Downsampled Asians (red)

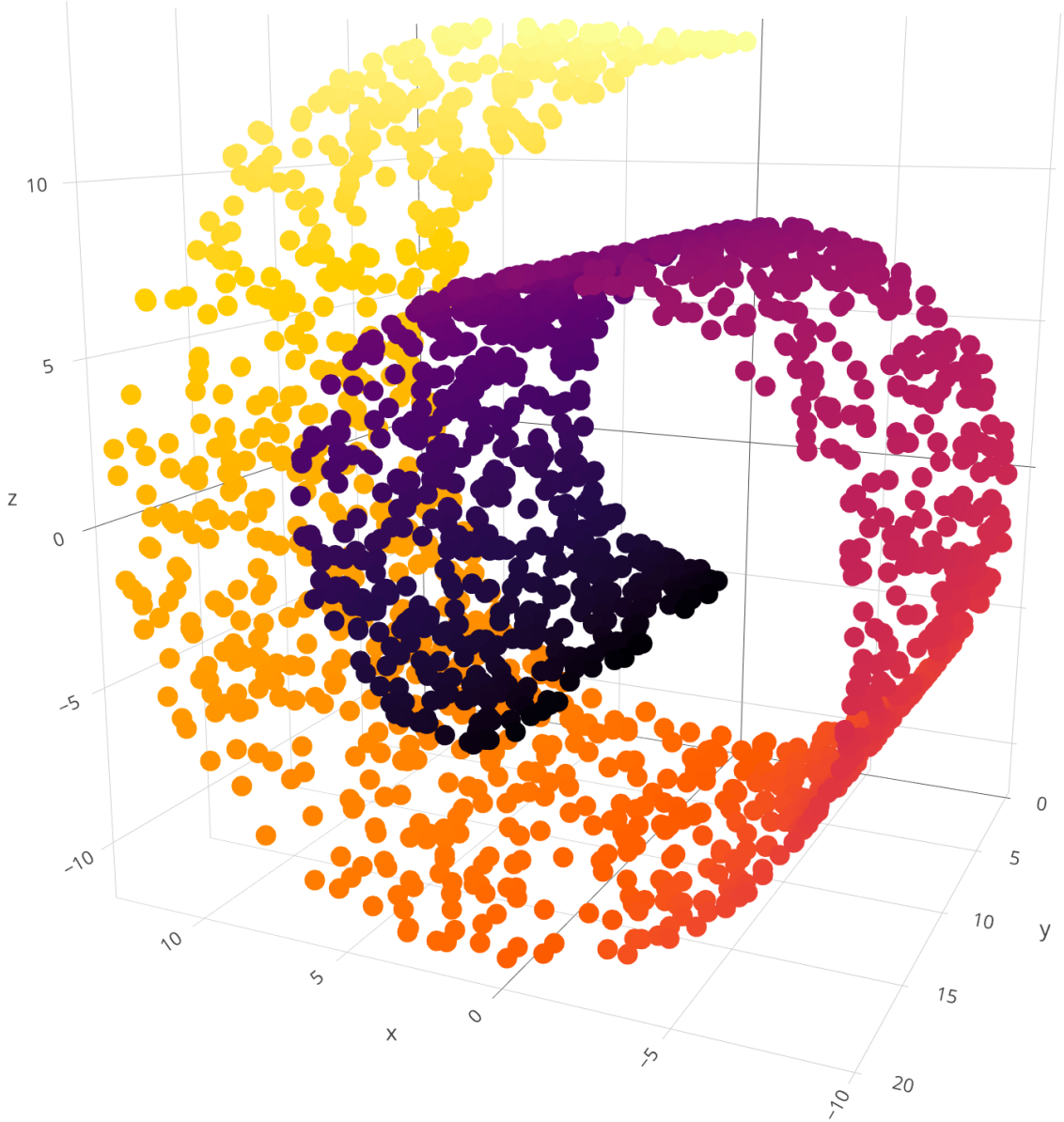
Slide courtesy Nikolay Oskolkov

Summary PCA & PLS-DA

- Linear multivariate methods of dimensionality reduction
- Uses components that aggregate observable variables (e.g. genes) in a model to summarize sources of variation in the data
- Interpretable/parametric dimensionality reduction

- PCA → unsupervised exploratory analysis (no information about sample groups included in the model)
- PLS-DA → supervised analysis to separate/discriminate sample

Non-linear dimensionality reduction


10
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x
0
−5
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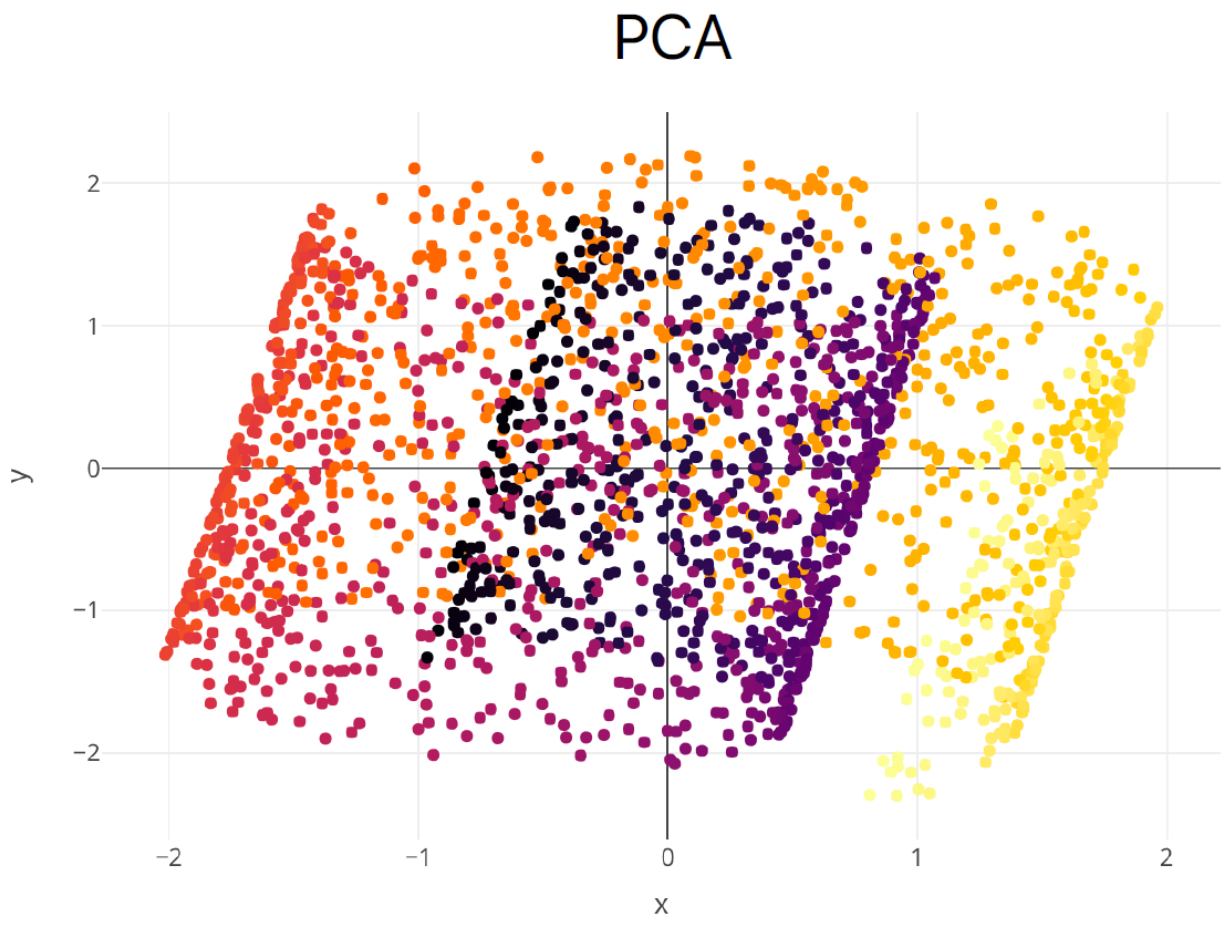
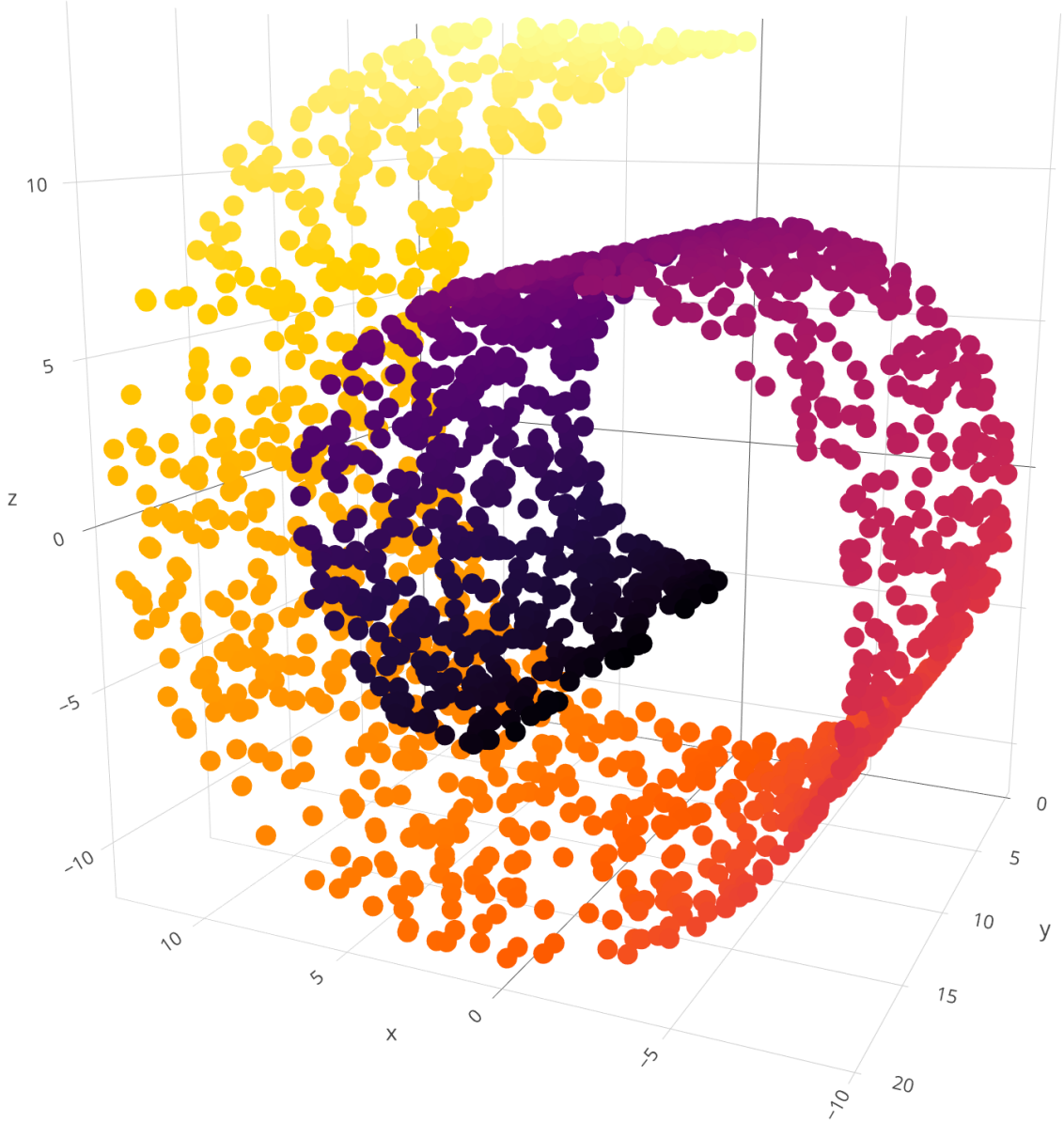
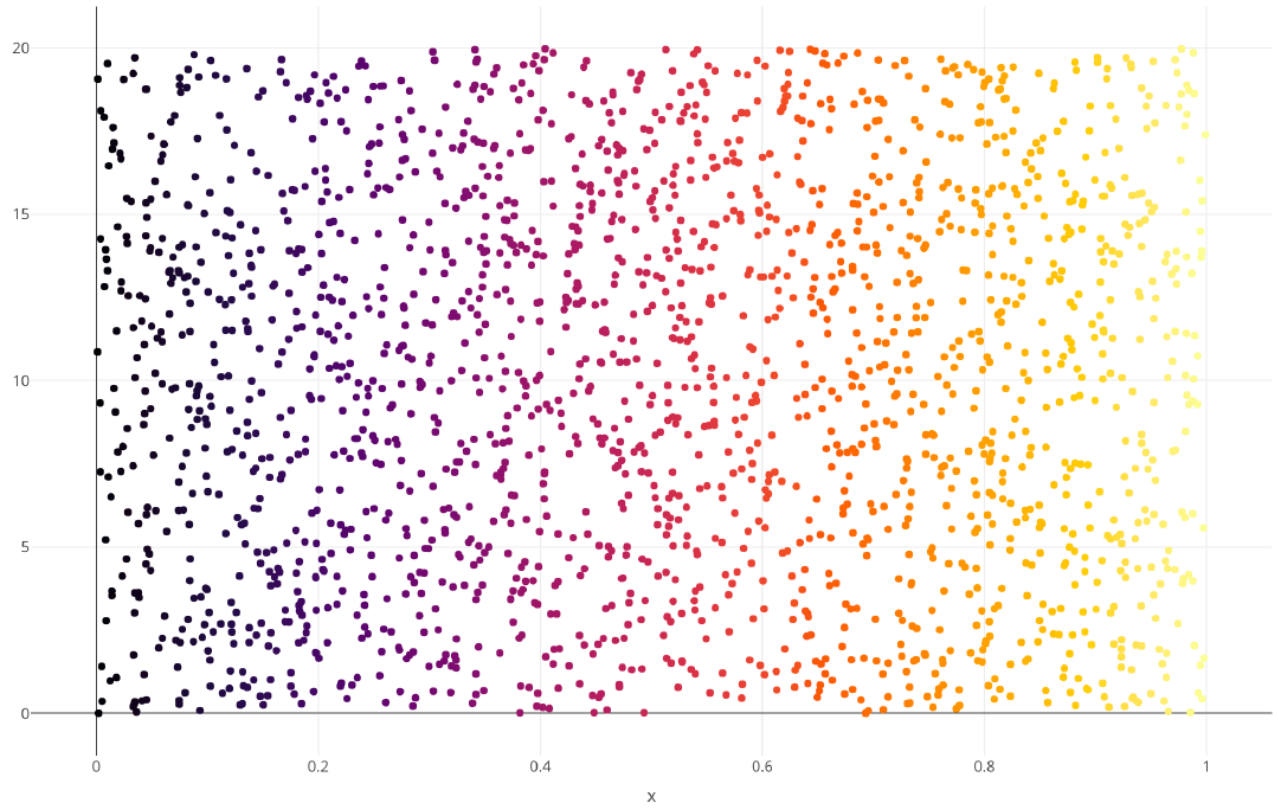
10
y
15
20


PCA
2
1
0
−1
−2
y
−2
−1
0
1
2
x

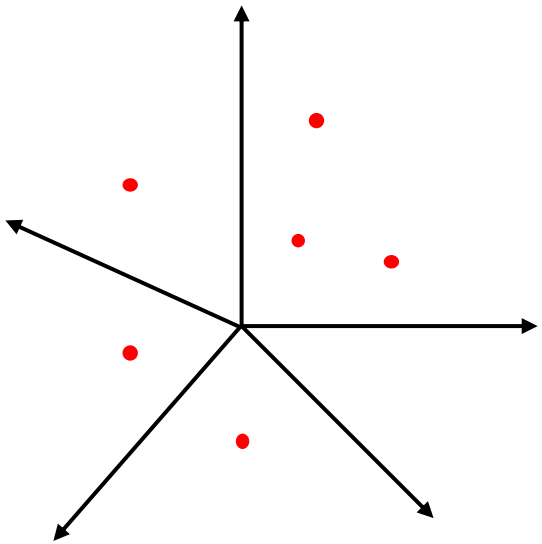


Better

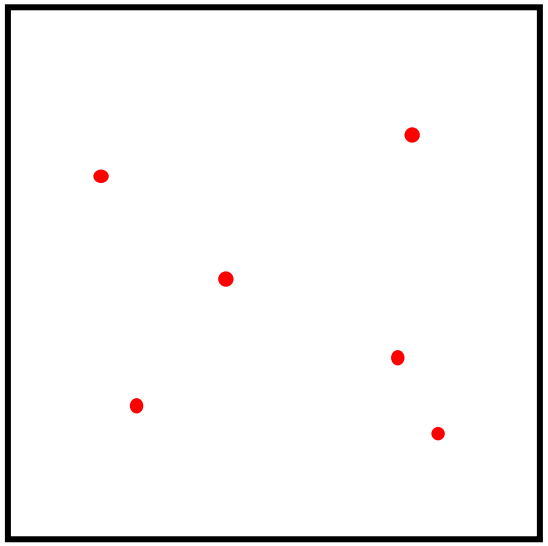


I. Multi-dimensional scaling (MDS)

P-dimensional space



MDS



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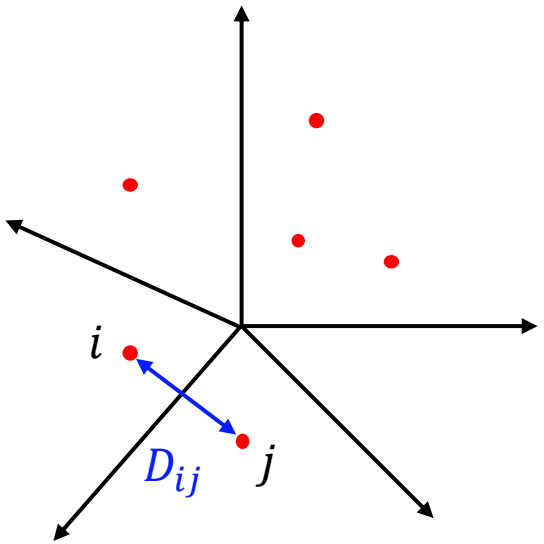
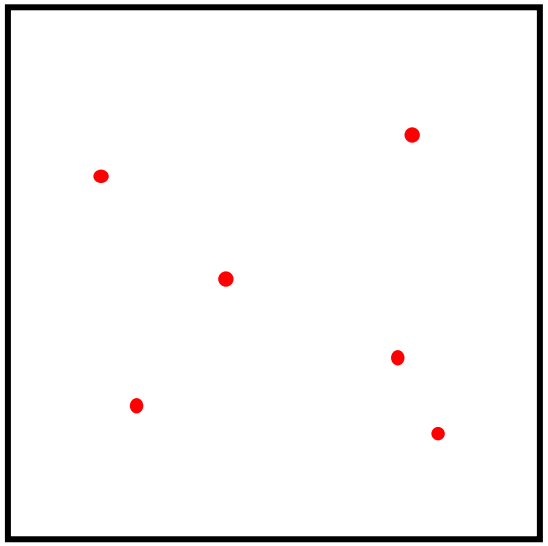


MDS



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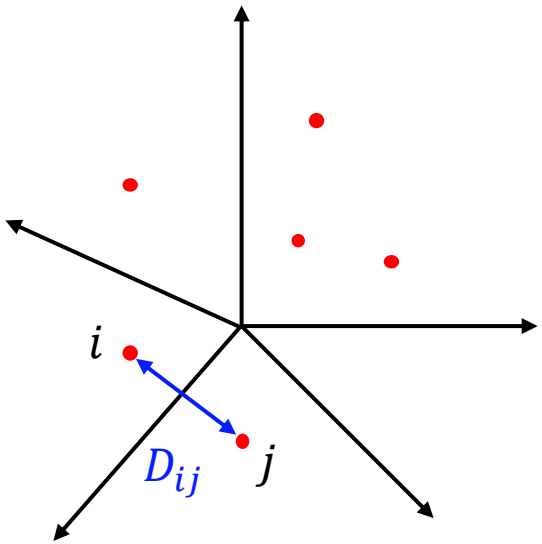
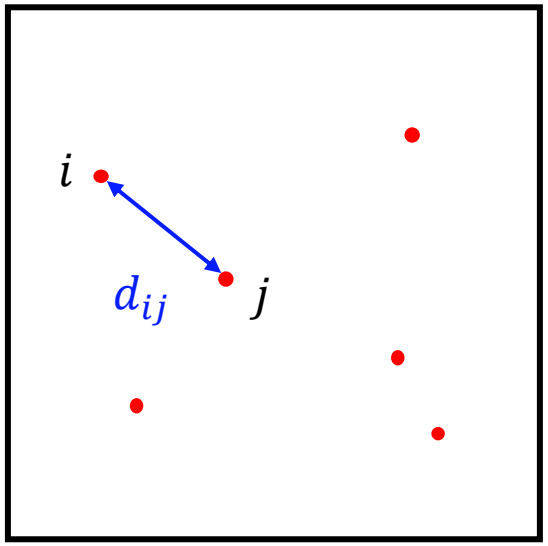
P-dimensional space



MDS



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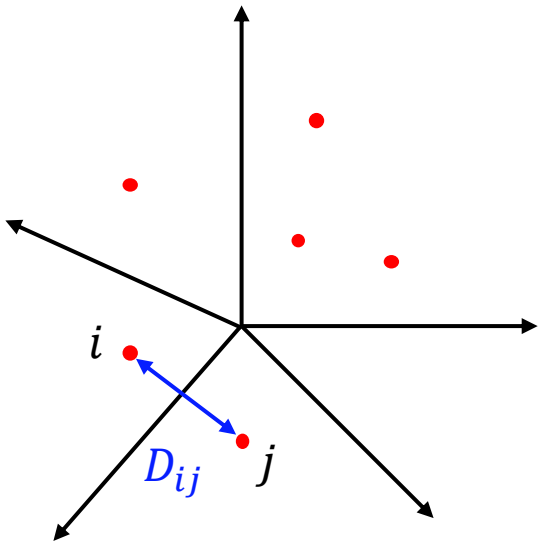
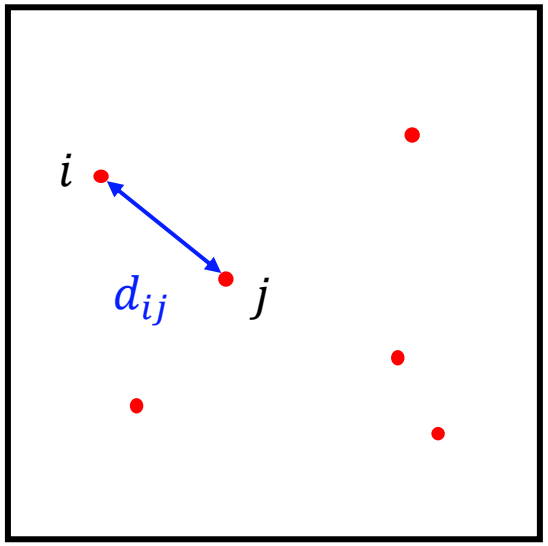


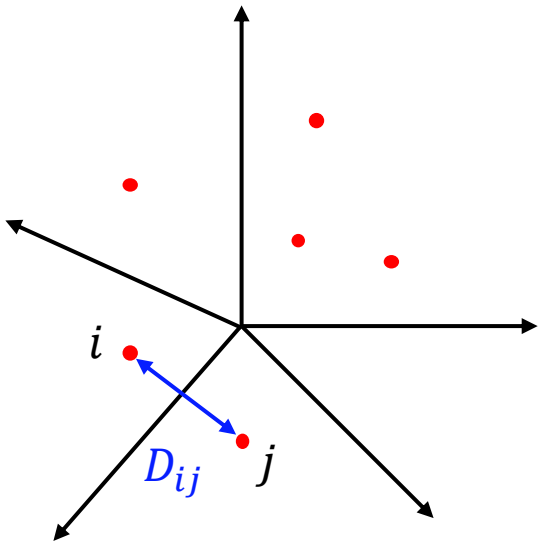
MDS



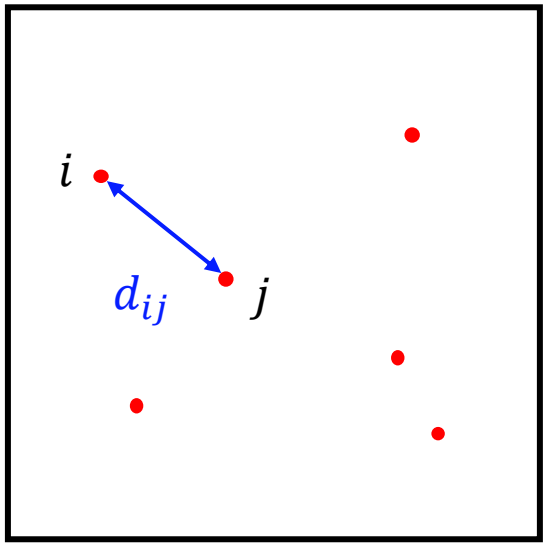
MDS objective: $\min \sum_{i,j} D_{ij} - d_{ij}$

I. Multi-dimensional scaling (MDS)

P-dimensional space



MDS



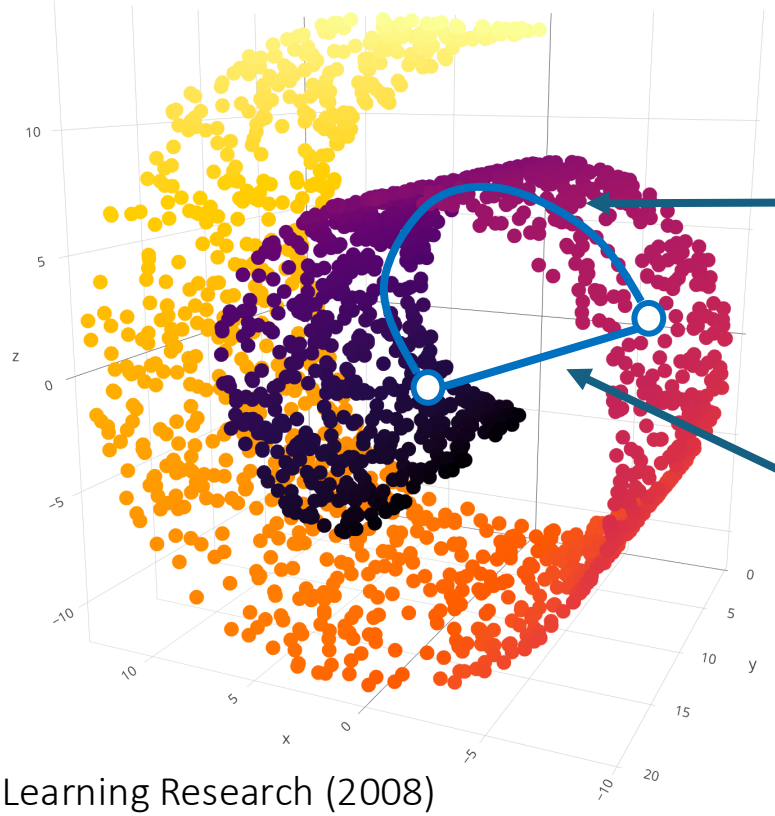
MDS objective: $\min \sum_{i,j} D_{ij} - d_{ij}$

Use gradient dissent to find the "best" embedding

II. t-SNE

t-distributed stochastic neighborhood embedding

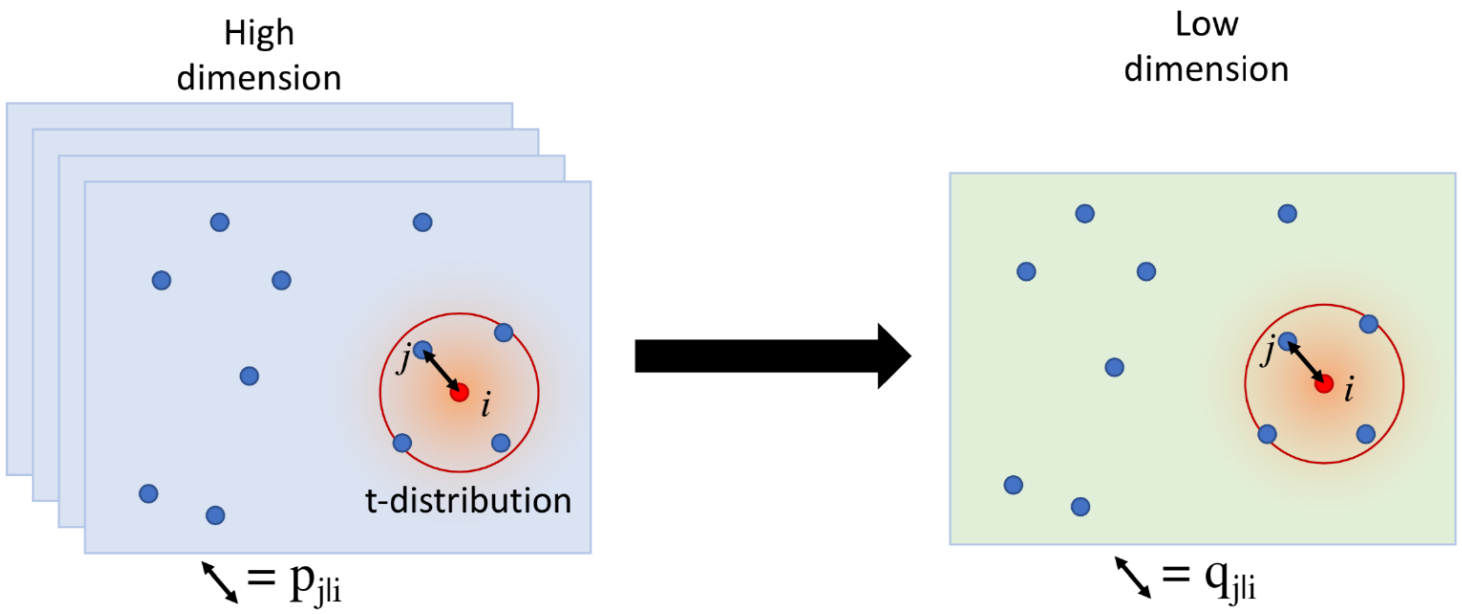
- Graph-based **non-linear** dimensionality reduction
- Calculated distances based on the distance to the **neighboring** cells



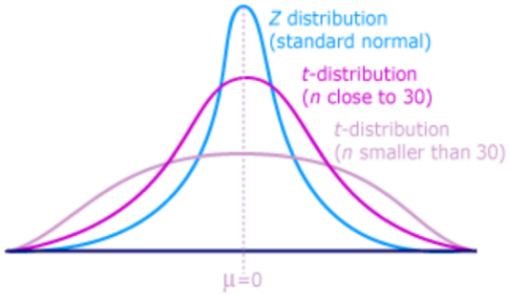
van der Maaten and Hinton, Journal of Machine Learning Research (2008)

t-SNE

t-distributed stochastic neighborhood embedding

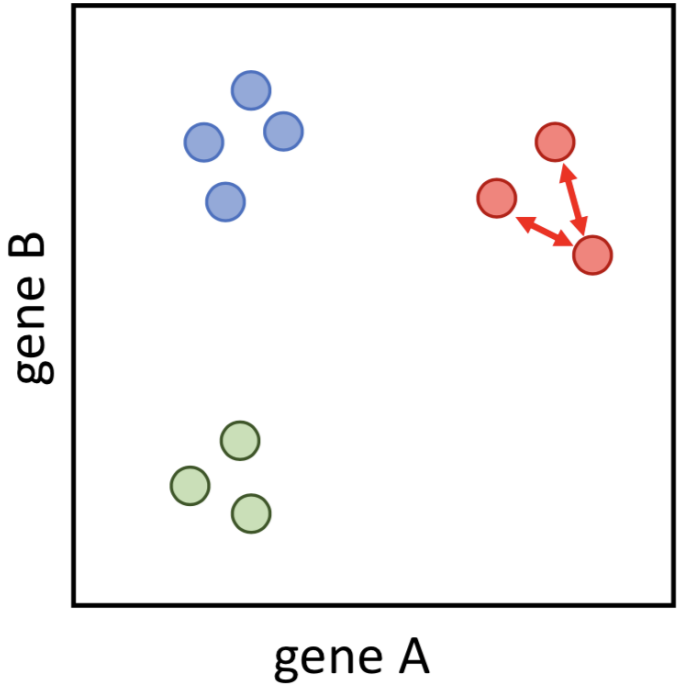


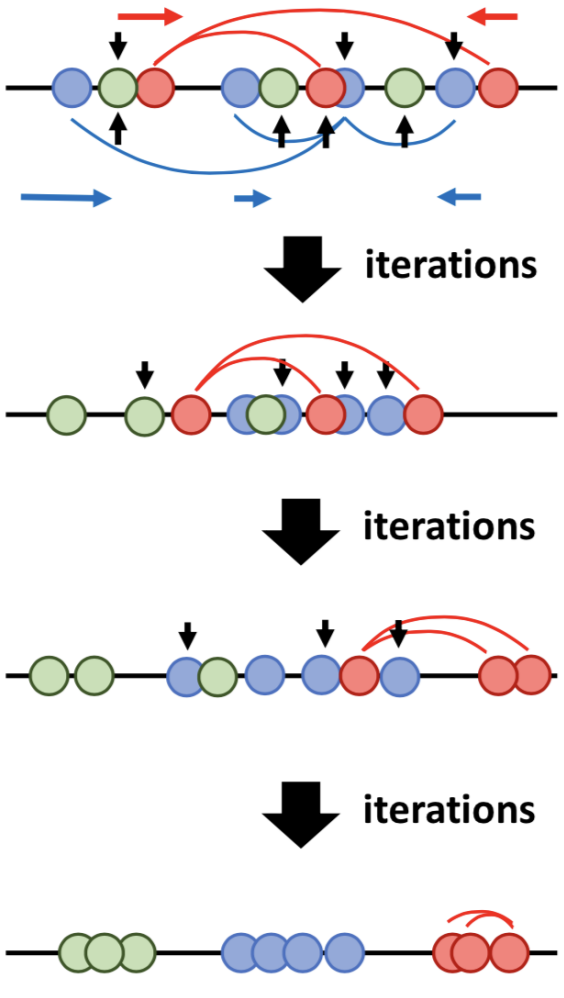
$p_{j|i}$ and $q_{j|i}$ measure the <u>conditional probability</u> that a point i would pick point j as it's nearest neighbor, in high (p) and low (q) dimensional space respectively.

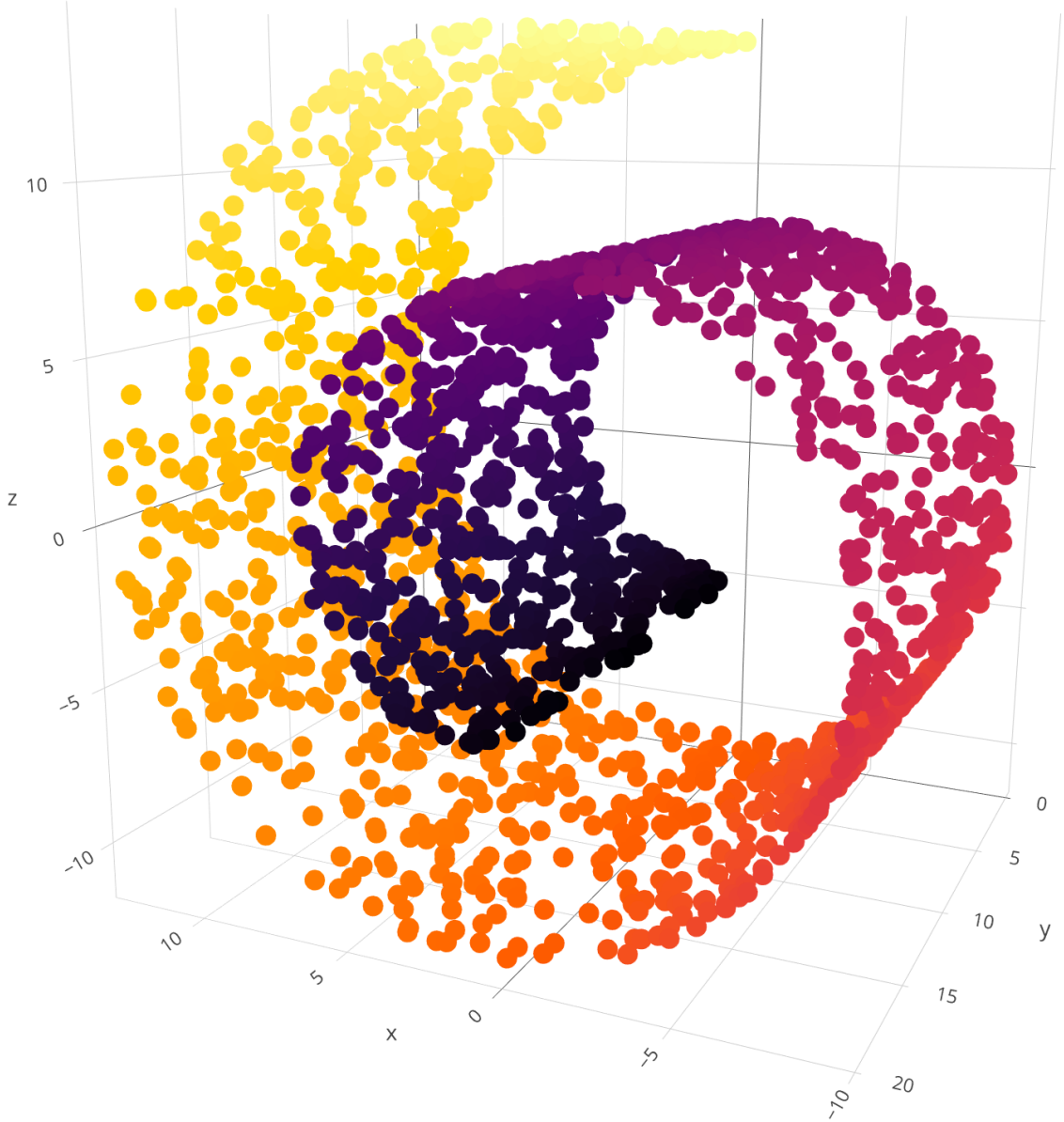


van der Maaten and Hinton, Journal of Machine Learning Research (2008)

t-SNE in brief






10
5
z
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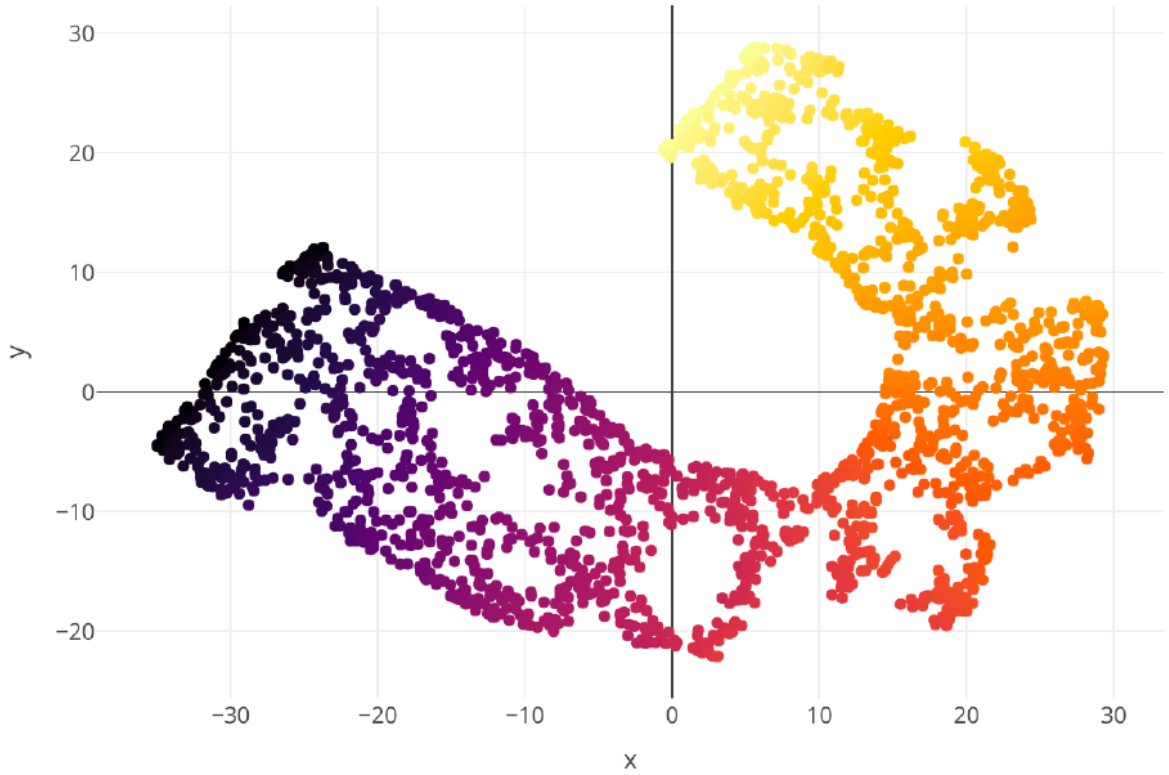
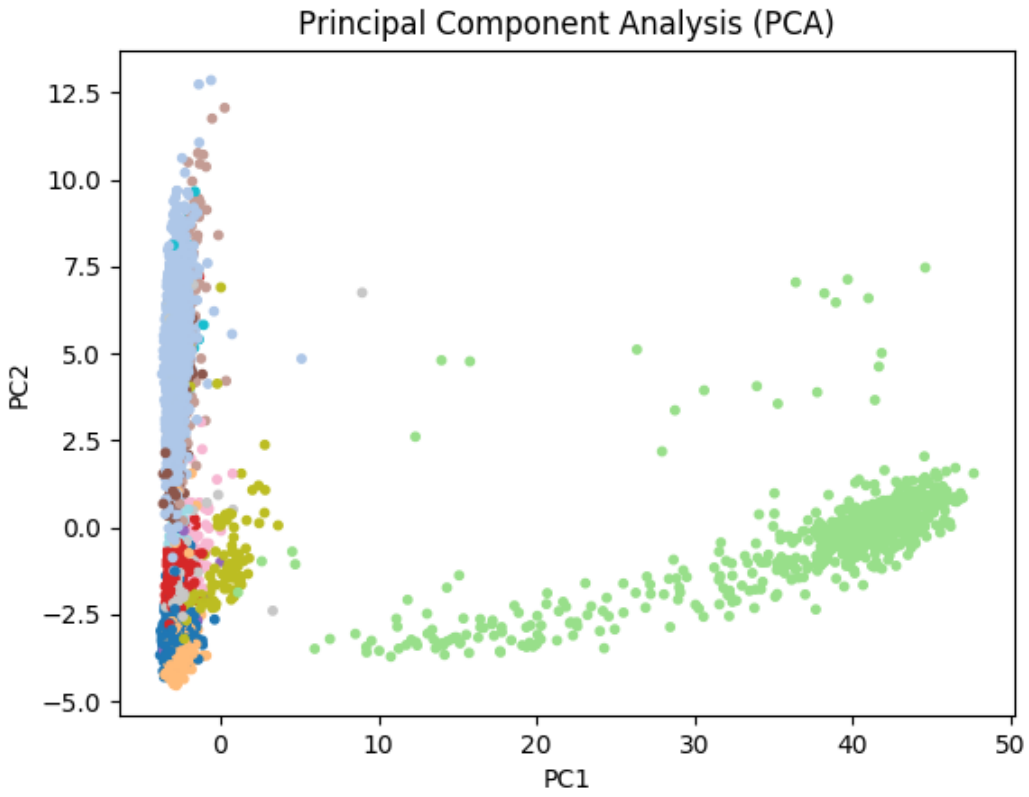
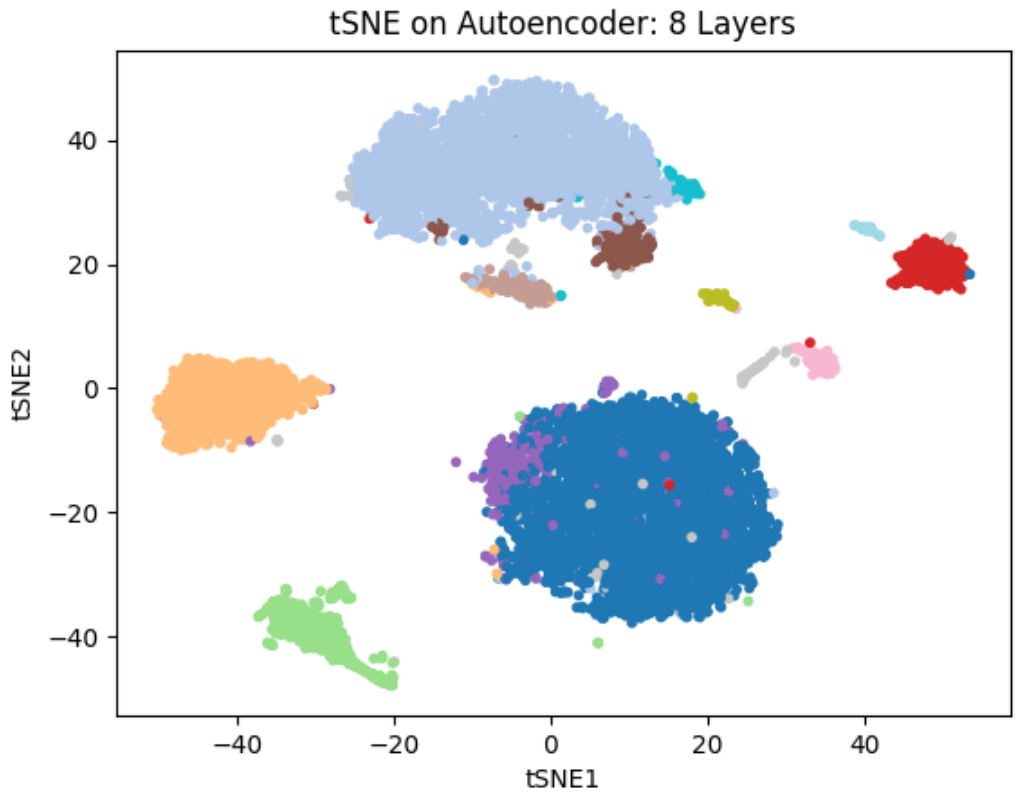
−10
10
5
x
0
−5
−10
0
5
10
y
15
20

t-SNE


30
20
10
y
0
−10
−20
−30
−20
−10
0
10
20
30
x

PCA vs tSNE for single cell data visualization





3k Peripheral Blood Mononuclear Cells (PBMC) available from 10X Genomics

Slide courtesy Nikolay Oskolkov

t-SNE

t-distributed stochastic neighborhood embedding

- **Non-linear** method of dimensionality reduction
- Can be run from the top PCs (e.g.: PC1 to PC10)
- Used for visualization, **not** for downstream analysis

<u>Problems</u>

- It does not learn an explicit function to map new points
- Cost function is not convex – This means that the optimal t-SNE cannot be computed
- Many hyper-parameters need to be defined empirically (dataset-specific)

Summary

- Dimensionality reduction techniques are very important for biological data exploration
- Linear (PCA and PLS-DA) vs non-linear methods (MDS and tSNE)
- Supervised (PLS-DA) vs unsupervised (PCA, MDS, and tSNE) approaches